

HOW TO LIVE IN A WORLD OF UNCERTAINTY

A practical introduction to Bayesian Inference

VITAL Training School
March 20th, 2025

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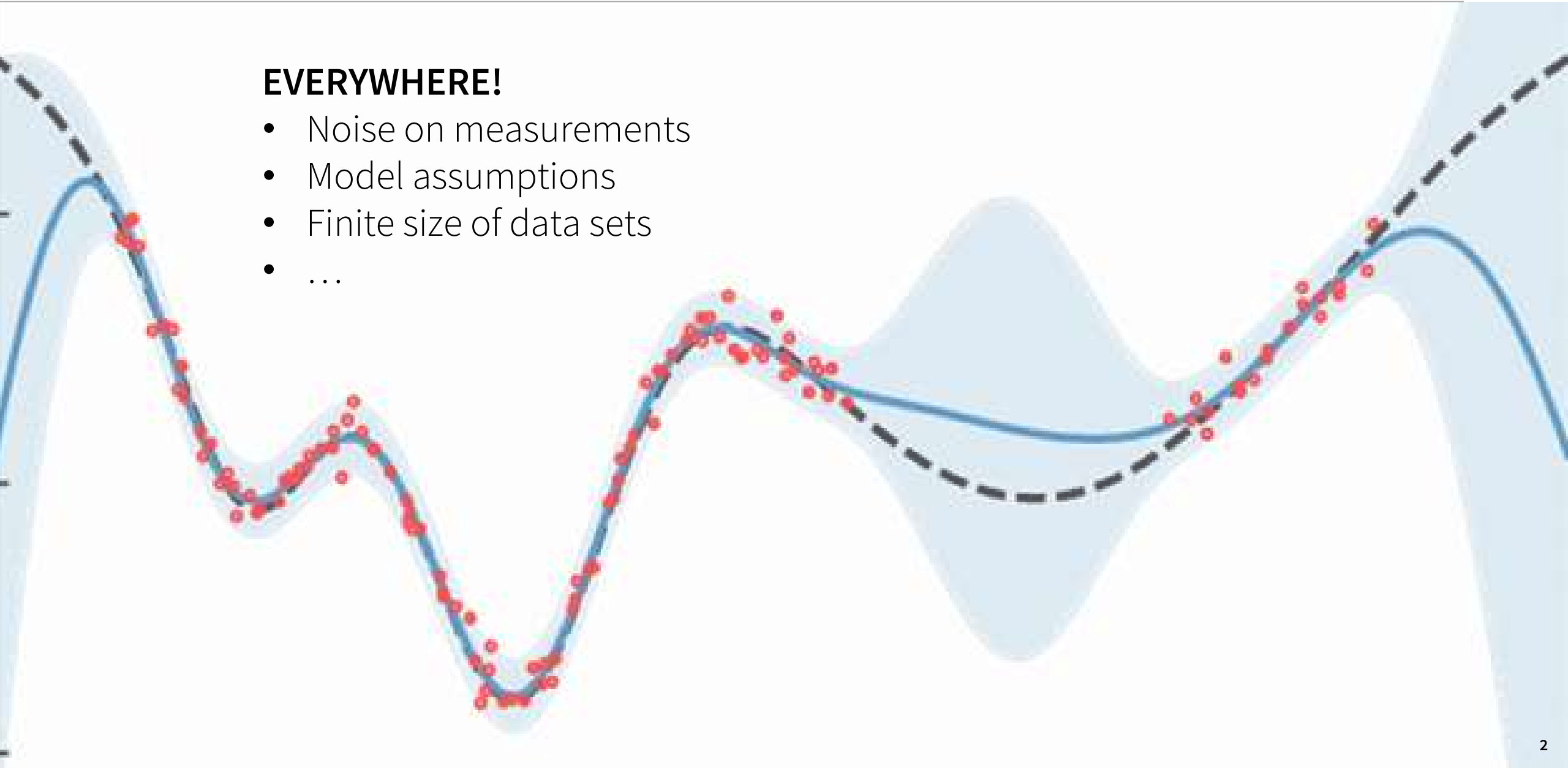


Co-funded by
the European Union

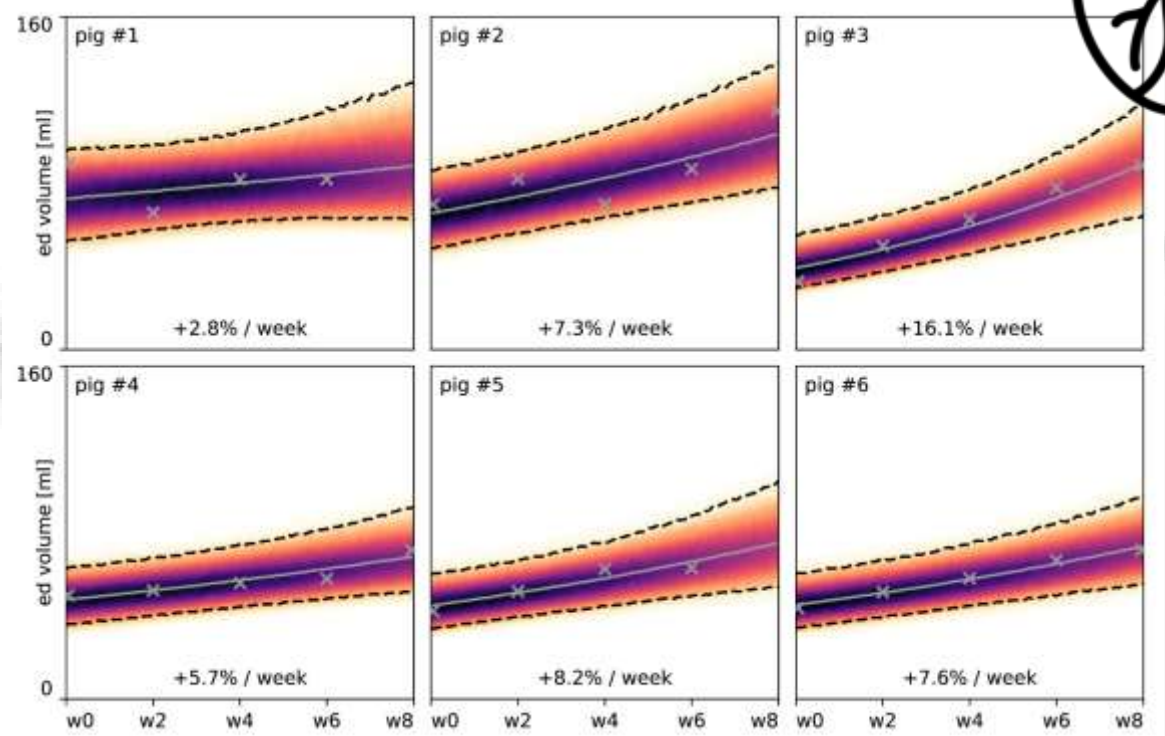
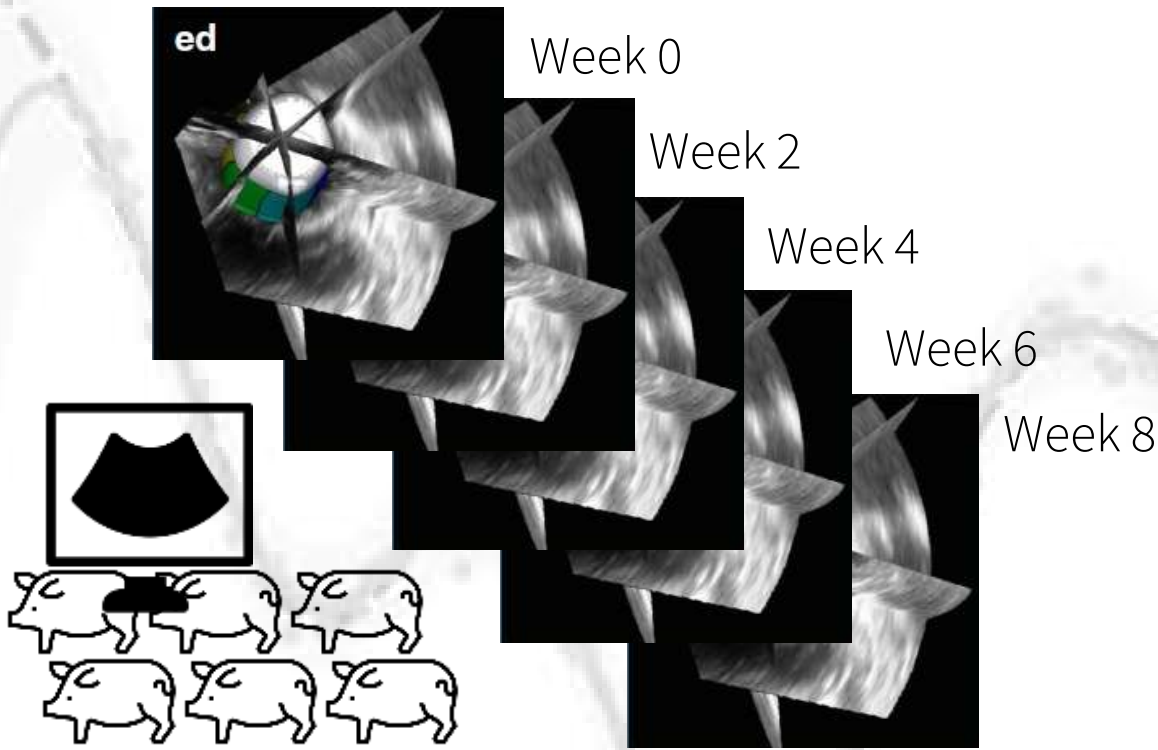
Uncertainty is ...

EVERYWHERE!

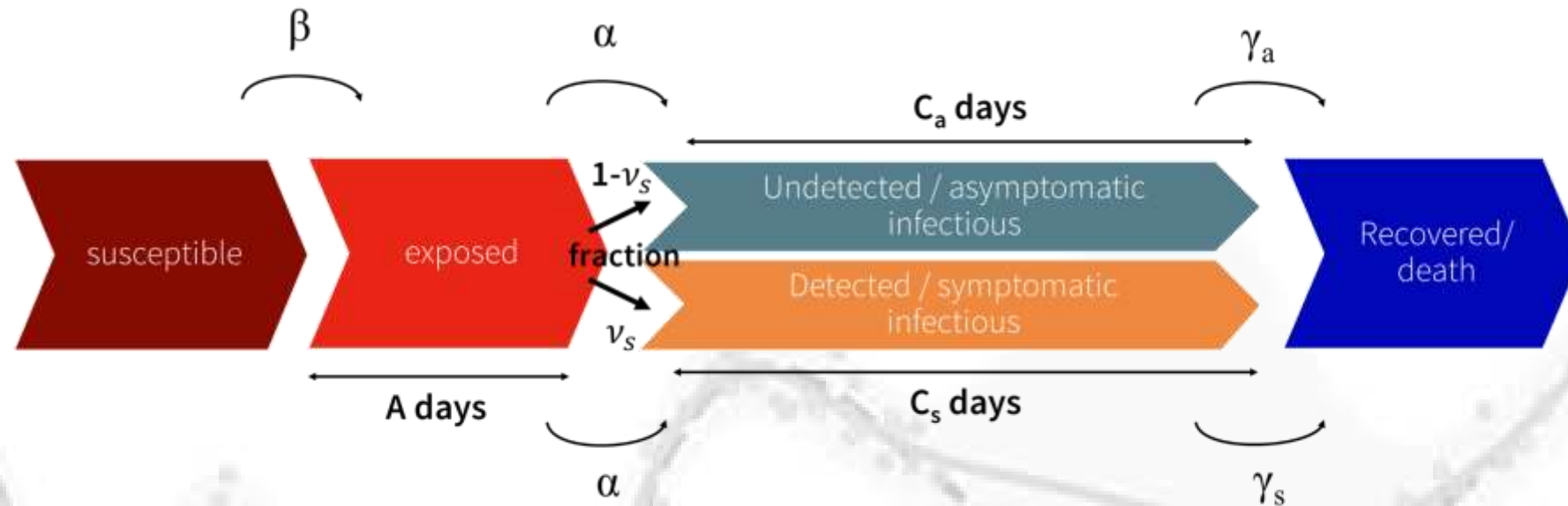
- Noise on measurements
- Model assumptions
- Finite size of data sets
- ...



Uncertainty is everywhere



Uncertainty is everywhere



$$\dot{S} = -\beta(t)SI$$

$$\dot{E} = \beta(t)SI - \alpha E$$

$$\dot{I}_s = v_s \alpha E - \gamma_s I_s$$

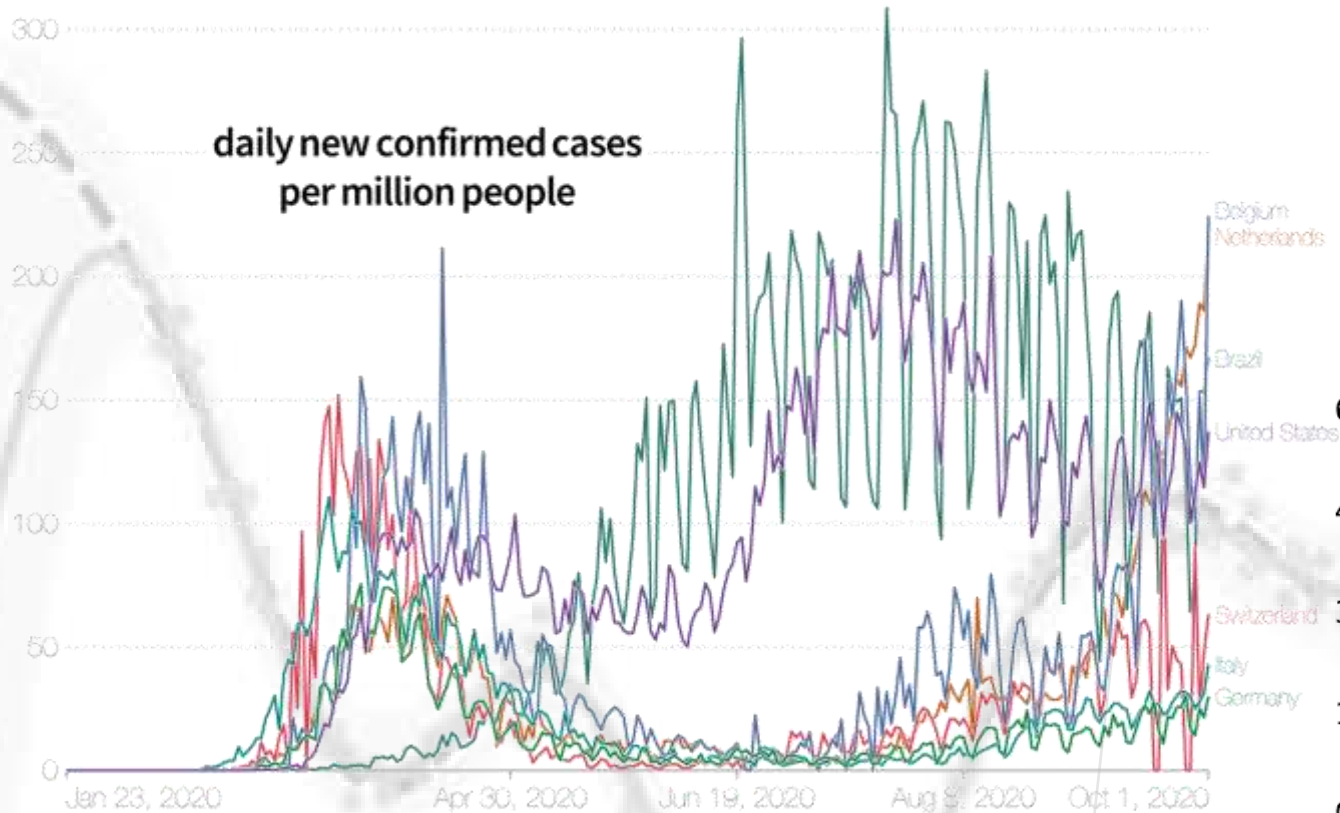
$$\dot{I}_a = (1 - v_s) \alpha E - \gamma_a I_a$$

$$\dot{R} = \gamma_s I_s + \gamma_a I_a$$

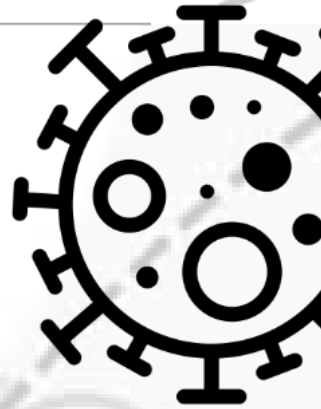
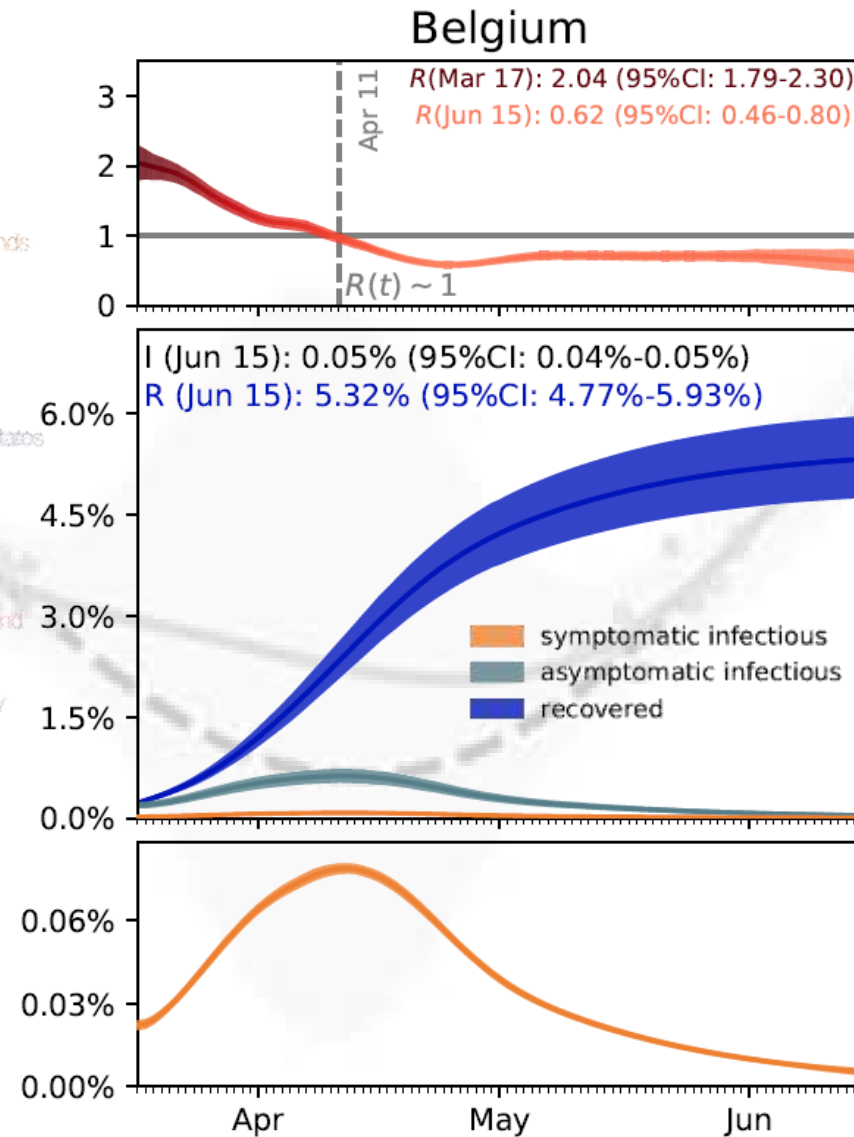
set of parameters

$$\boldsymbol{\vartheta} = \{A, C_s, C_a, v_s, \beta(t), E_0, I_{a0}, I_{s0}, \boldsymbol{\sigma}\}$$

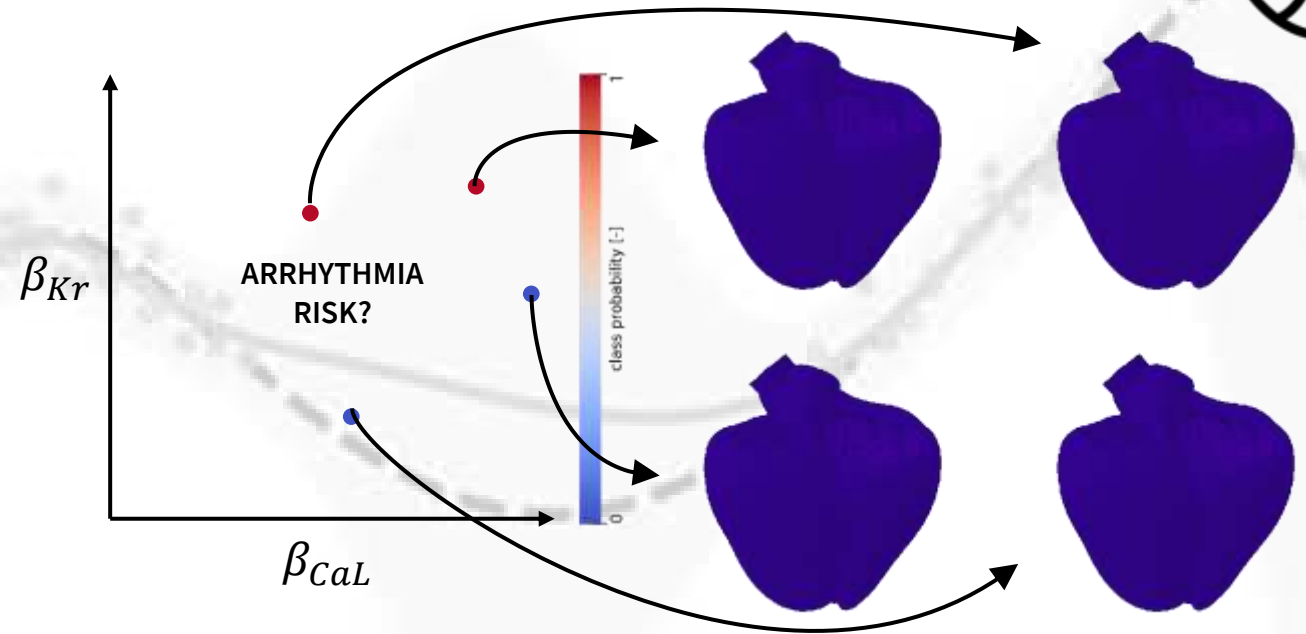
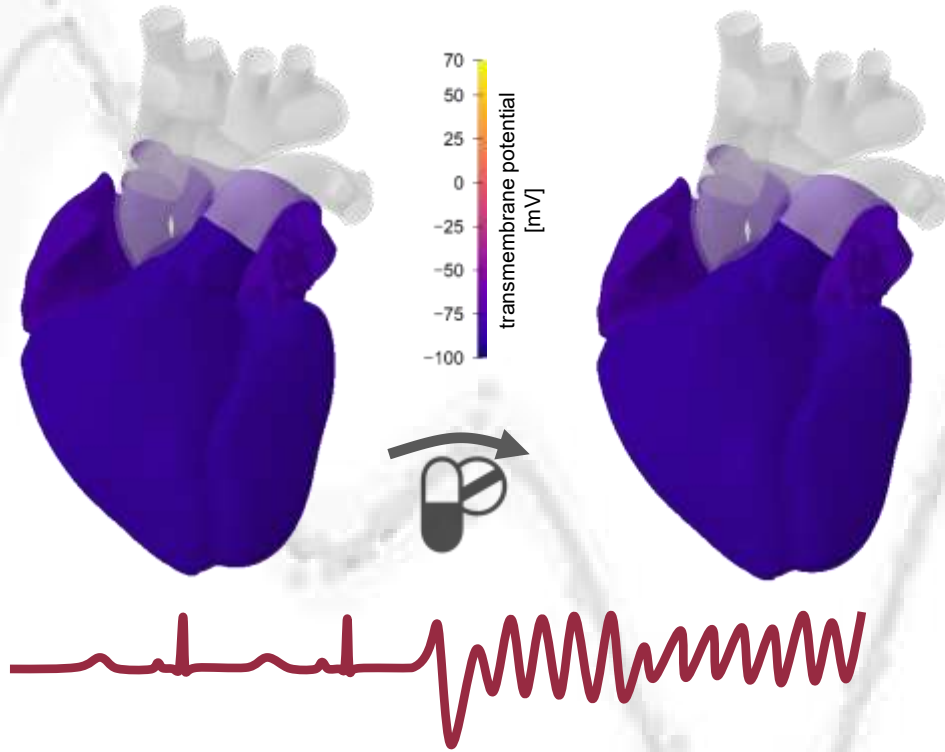
Uncertainty is everywhere



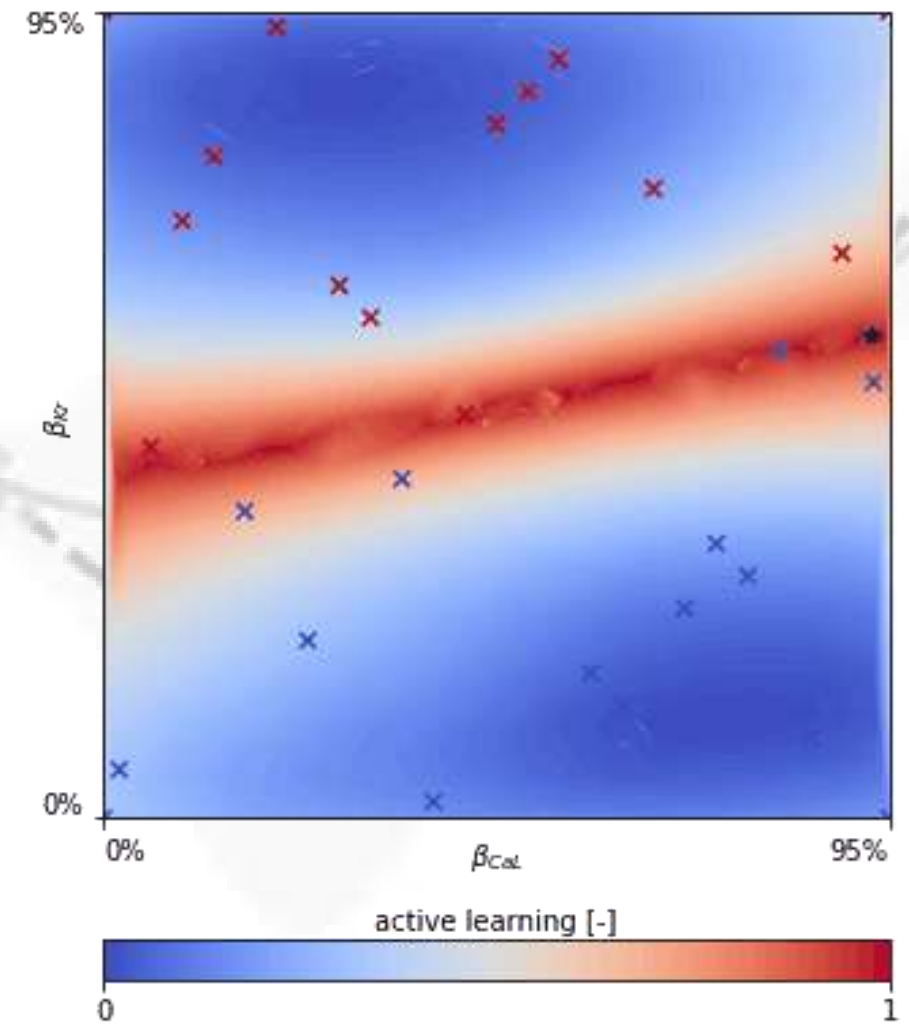
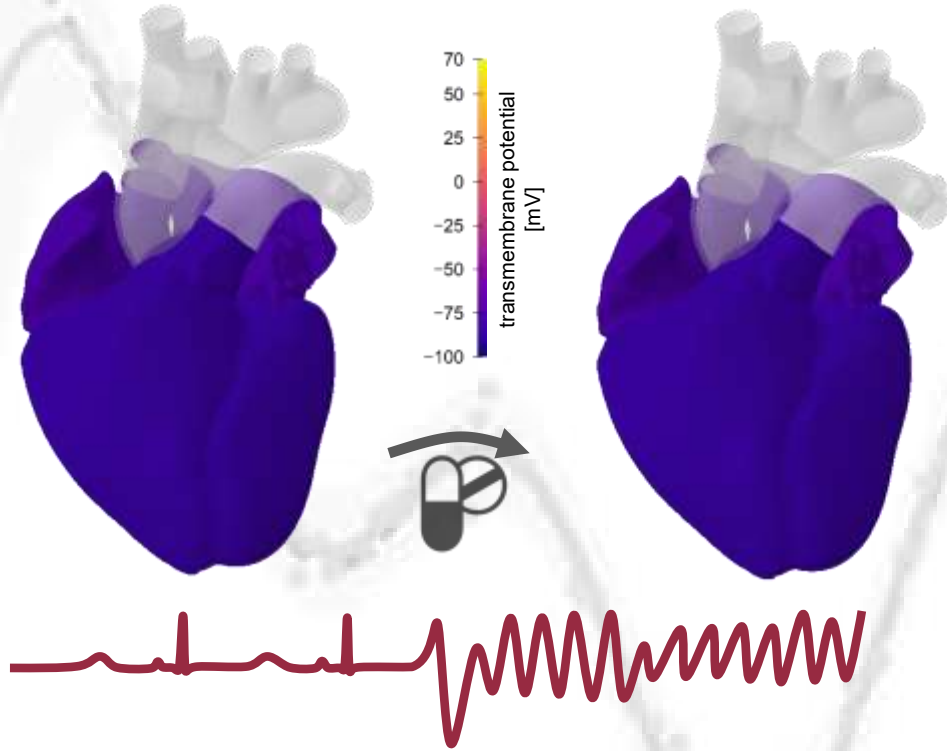
Belgium
April 14 – May 13, 2020
undercount: 8-12x



Uncertainty is everywhere



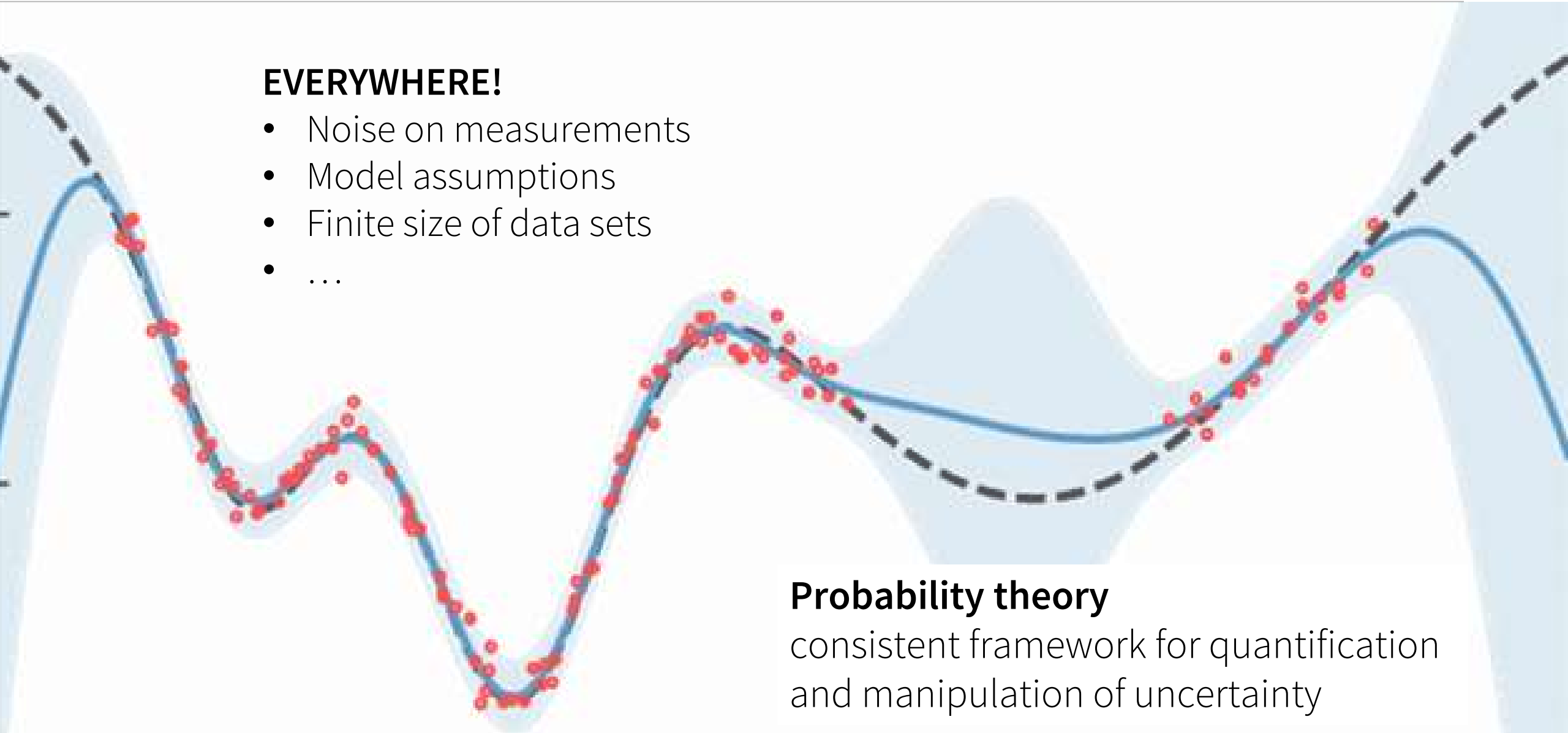
Uncertainty is everywhere



Uncertainty is ...

EVERYWHERE!

- Noise on measurements
- Model assumptions
- Finite size of data sets
- ...



Probability theory

consistent framework for quantification
and manipulation of uncertainty

Bayes' theorem



Thomas Bayes [1701-1761]
English statistician and philosopher

Bayes' theorem

gives us a calculation for updating our beliefs

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

prior
belief



posterior
belief

Where there's smoke...

Dangerous fires are rare (1 percent), but since you live next to a Peruvian chicken restaurant, smoke is fairly common (20 percent). Almost all fires come with smoke (95 percent). You see smoke out of your window. Is it time to evacuate?

$$P(\text{Fire} | \text{Smoke}) = \frac{\overset{\text{Probability of dangerous fire}}{P(\text{Fire})} \overset{\text{Probability of seeing smoke if there is a fire}}{P(\text{Smoke} | \text{Fire})}}{\underset{\text{Probability of seeing smoke}}{P(\text{Smoke})}}$$

Probability that there is a dangerous fire when you see smoke

**WE ARE NOT WORKING
WITH EVENTS,
WE WORK WITH DATA**

Statistics is the study of data

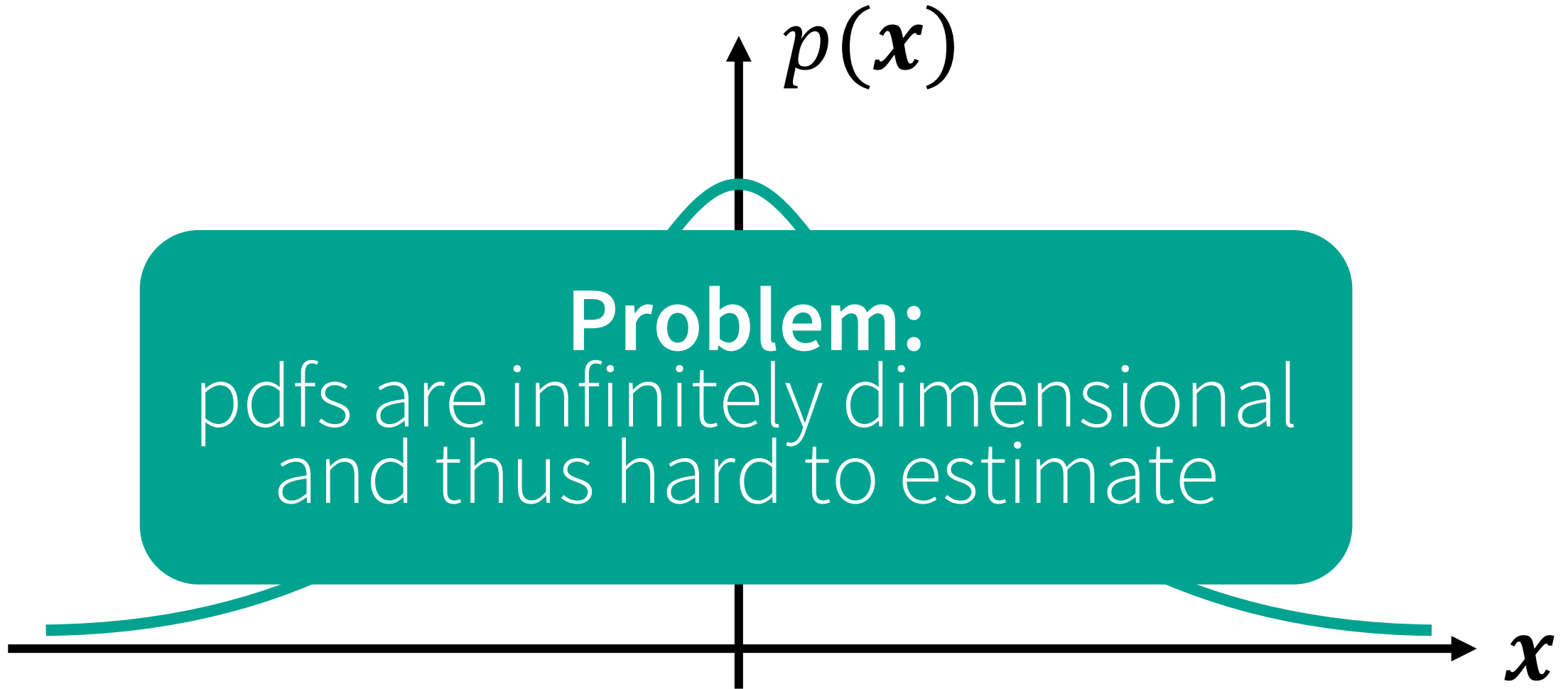
$\boldsymbol{x} \longrightarrow$ data (random in nature)

We assume this randomness has a predictive form or structure, i.e.

$p(\boldsymbol{x})$

probability density function

Statistics is the study of data

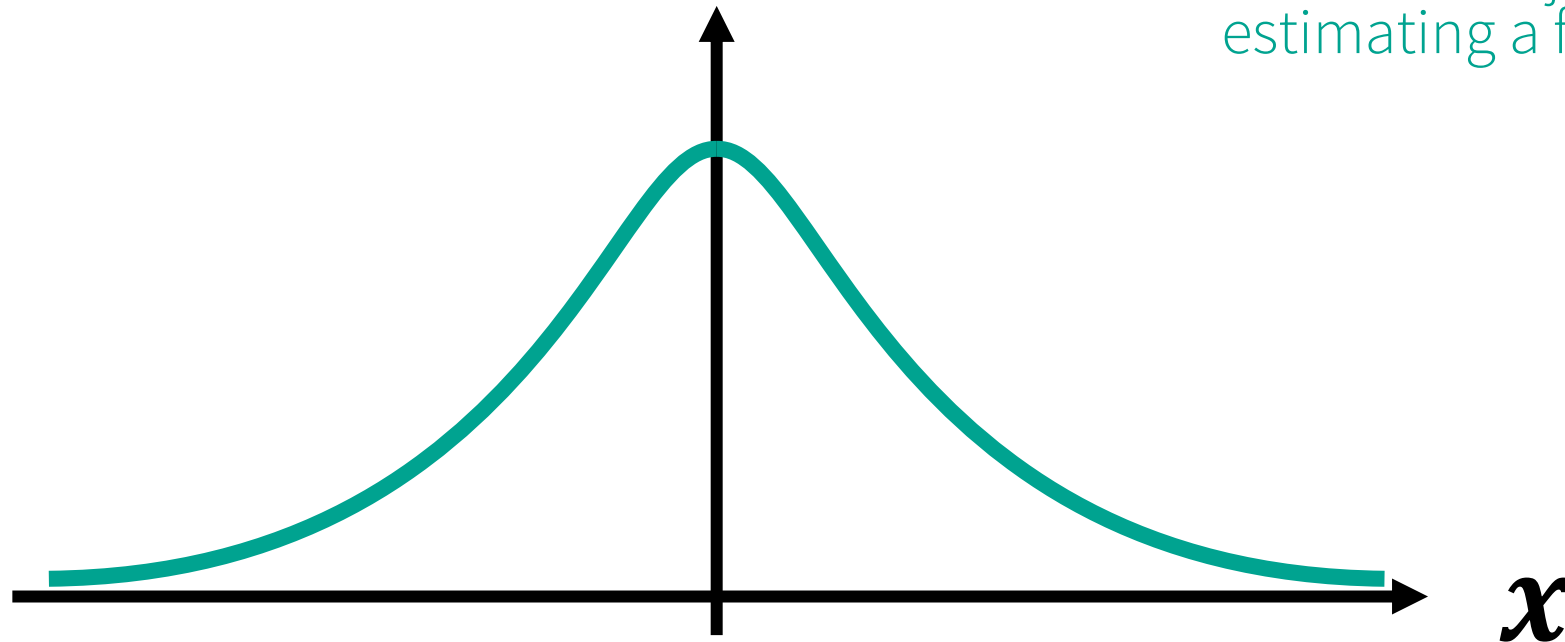


Statistics is the study of data

Strategy:
approximate pdfs as
“parametric family”

$$p(x) \longrightarrow p_{\theta}(x)$$

You can get an entire
function just from
estimating a few values θ



Statistics is the study of data

$$p_{\theta}(x)$$

two philosophies $\rightarrow \theta$

FREQUENTISTS

Probability = frequency

The one you learned in uni

BAYESIANS

Probability = degree of belief

Bayes' theorem

FREQUENTISTS

Probability = frequency

θ is fixed and unknown

we can estimate θ through
maximum likelihood estimation

$$\hat{\theta}_{MLE} \rightarrow \theta$$

hypothesis testing

$$H_0: \theta = \theta_0$$

$$P(|T| \geq t | \theta = \theta_0)$$

*The data is unlikely to have come from this
specific (null) parameter value*

p-value

Two philosophies



$$P(|T| \geq t | \theta = \theta_0)$$

FREQUENTISTS

Probability = frequency

BAYESIANS

Probability = degree of belief

Two philosophies



$$P(\theta \mid \text{data } \mathbf{x})$$

FREQUENTISTS

Probability = frequency

BAYESIANS

Probability = degree of belief

Two philosophies



$$P(\theta \mid \text{data } \mathbf{x})$$

BAYESIANS

Probability = degree of belief

θ is a random value

we can have prior beliefs/knowledge
about the parameter θ

we update our beliefs
based on the data $\mathbf{x}$

Bayesian philosophy

Bayes' theorem

describes the probability of a hypothesis/parameters, based on prior knowledge and observed evidence

$$P(\theta|x) = \frac{P(x|\theta)P(\theta)}{P(x)}$$

$P(\theta|x)$ **POSTERIOR:** probability of the hypothesis/parameter θ given the evidence x

$P(\theta)$ **PRIOR:** our initial beliefs about the hypothesis/parameter θ

$P(x|\theta)$ **LIKELIHOOD:** probability of the evidence x given hypothesis/parameter θ

$P(x)$ **EVIDENCE:** probability of the evidence x arising, globally

Thomas Bayes [1701-1761]
English statistician and philosopher



FREQUENTIST VS BAYESIAN

The coin tossing example

Central question:

What is the probability that the coin will land heads up?

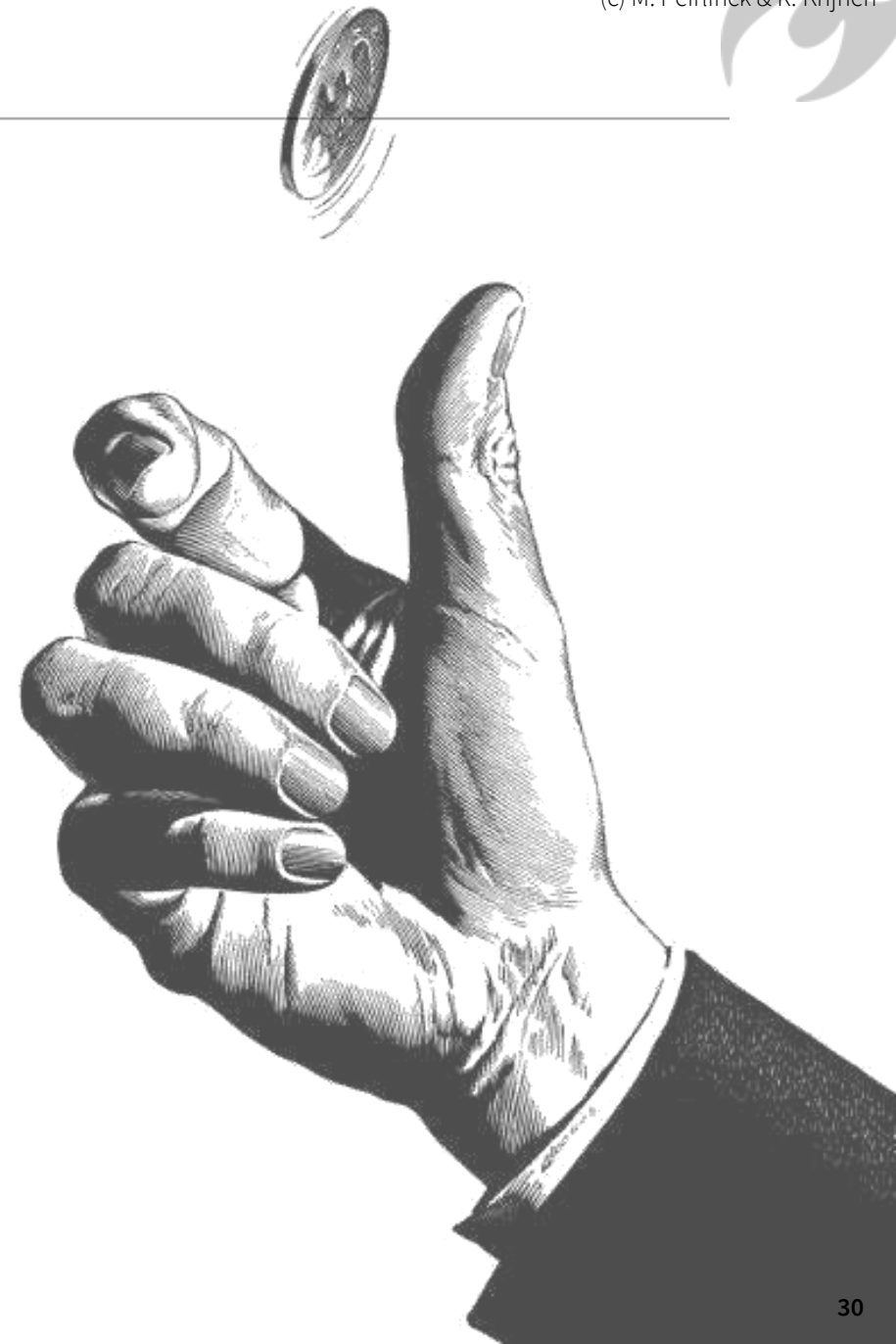
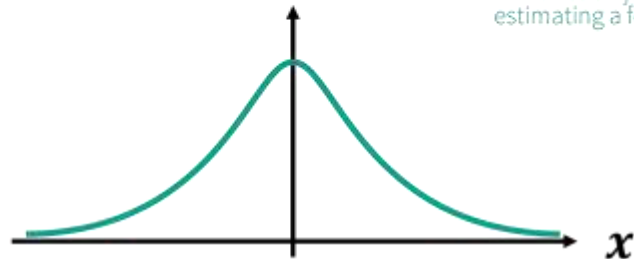
heads: $x = 1$

tails: $x = 0$

Strategy:
approximate pdfs as
“parametric family”

$$p(x) \longrightarrow p_{\theta}(x)$$

You can get an entire
function just from
estimating a few values θ



The coin tossing example

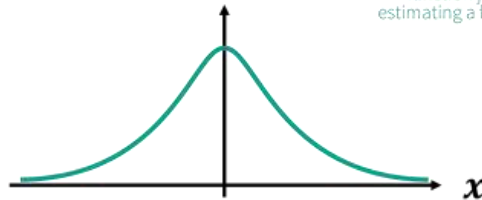
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Strategy:
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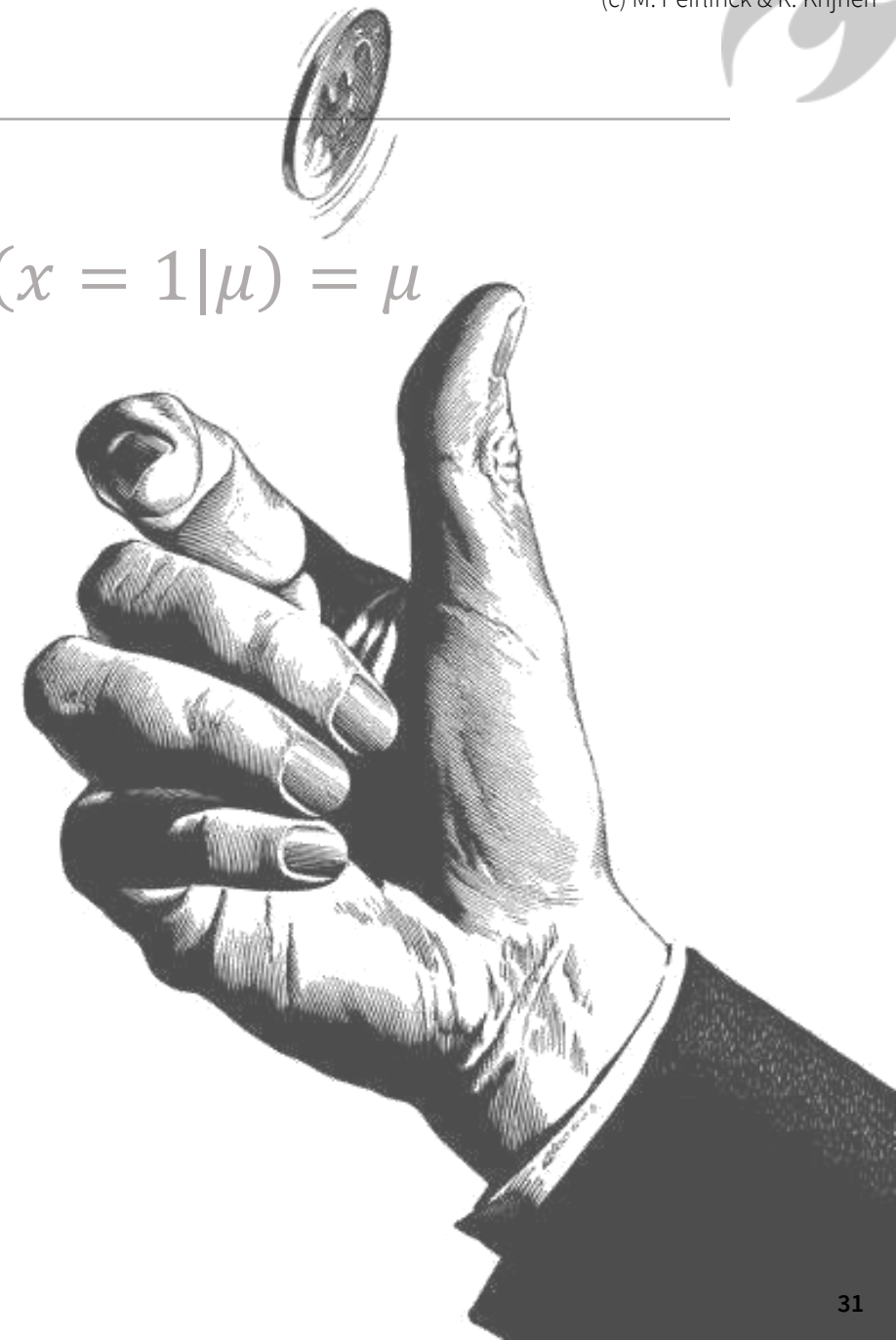
You can get an entire
function just from
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$$\begin{aligned} p(x|\mu) &= \text{Bern}(x|\mu) \\ &= \mu^x (1 - \mu)^{1-x} \end{aligned}$$

Jacob Bernouilli [1654-1705]
Swiss mathematician

$$p(x = 1|\mu) = \mu$$



The coin tossing example

$$p(x|\mu) = \text{Bern}(x|\mu) = \mu^x(1 - \mu)^{1-x}$$

A sequence of n i.i.d. coin flips

= a *Bernoulli* process of
 $D = \{x_1, x_2, \dots, x_n\}$ random variables

probability of m successes in n trials

$$\begin{aligned} p(x_i = m) &= \text{Bin}(m|n, \mu) \\ &= \binom{n}{m} \mu^m (1 - \mu)^{n-m} \end{aligned}$$

Jacob Bernoulli [1654-1705]
Swiss mathematician



The coin tossing example

$$p(x_i = m) = \binom{n}{m} \mu^m (1 - \mu)^{n-m}$$

given a dataset

$$D = \{x_1, x_2, \dots, x_n\}$$

What is μ ?

FREQUENTISTS

Probability = frequency

BAYESIANS

Probability = degree of belief



The coin tossing example: the frequentist approach

Maximizing the likelihood of the data D
given the parameter μ :

$$p(\mathcal{D} \mid \mu) = \prod_{i=1}^n p(x_i \mid \mu) = \prod_{i=1}^n \binom{n}{m} \mu^{x_i} (1 - \mu)^{1-x_i} = \binom{n}{m} \mu^m (1 - \mu)^{n-m}$$

$$\log p(\mathcal{D} \mid \mu) = \log \binom{n}{m} + m \log \mu + (n - m) \log(1 - \mu)$$

$$\begin{aligned} \frac{\partial}{\partial \mu} \log p(\mathcal{D} \mid \mu) &= \frac{\partial}{\partial \mu} \log \binom{n}{m} + \frac{\partial}{\partial \mu} m \log \mu + \frac{\partial}{\partial \mu} (n - m) \log(1 - \mu) \\ &= \frac{m}{\mu} - \frac{n - m}{1 - \mu} \end{aligned}$$



The coin tossing example: the frequentist approach

Maximizing the likelihood of the data D
given the parameter μ :

$$\begin{aligned}\frac{\partial}{\partial \mu} \log p(\mathcal{D} \mid \mu) &= \frac{\partial}{\partial \mu} \log \binom{n}{m} + \frac{\partial}{\partial \mu} m \log \mu + \frac{\partial}{\partial \mu} (n - m) \log(1 - \mu) \\ &= \frac{m}{\mu} - \frac{n - m}{1 - \mu}\end{aligned}$$

$$\mu_{ML} = \frac{m}{n}$$



The coin tossing example: the frequentist approach

Now assume:

- Fair coin (i.e. true $\mu = 0.5$)
- Three coin tosses, heads 3x in a row (happens $1/8$ times)

Central question:

What is the probability that the coin will land heads up?

$$p(x = 1|\mu) = \mu = \mu_{ML} = \frac{m}{n} = \frac{3}{3} = 1$$



The coin tossing example

$$p(x_i = m) = \binom{n}{m} \mu^m (1 - \mu)^{n-m}$$

given a dataset

$$D = \{x_1, x_2, \dots, x_n\}$$

What is μ ?

FREQUENTISTS

Probability = frequency

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The coin tossing example: the Bayesian approach

Rather than maximizing $\log p(D \mid \mu)$, we want to place a prior probability on μ , and thus maximize (maximum a posteriori):

$$\underbrace{\log p(D \mid \mu) + \log p(\mu)}_{\text{What prior to put on } \mu?}$$

log likelihood log prior

What prior to put on μ ?



The coin tossing example: the Bayesian approach

What prior to put on μ ?

Depends on the assumed/prior knowledge:

- › Assume the coin is fair,
i.e. $\mu = 0.5$ is mode of distribution
- › Assume that biased coins do not prefer heads over tails or vice versa,
i.e. distribution is symmetric.

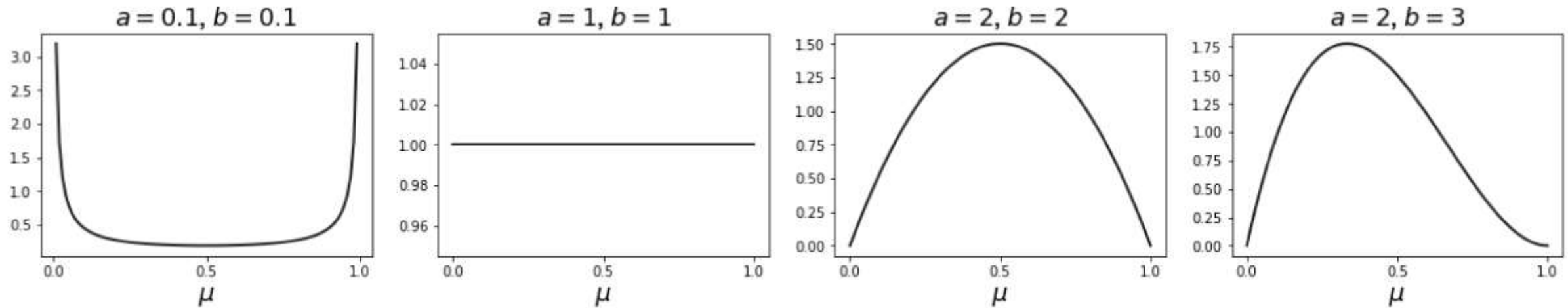
one option: the ***beta distribution***



The coin tossing example: the Bayesian approach

What prior to put on μ ?

one option: the **beta distribution**



$$\text{Beta}(\mu \mid a, b) = \frac{\Gamma(a + b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1 - \mu)^{b-1}$$

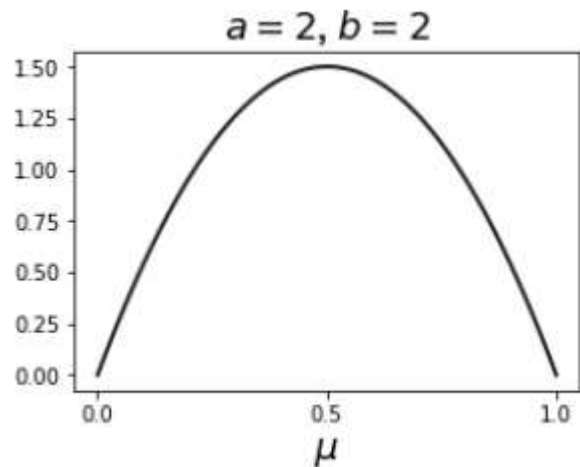
hyperparameters a and b control shape of distribution

The coin tossing example: the Bayesian approach

What prior to put on μ ?

one option: the **beta distribution**

- › Assume the coin is fair,
i.e. $\mu = 0.5$ is mode of distribution
- › Assume that biased coins do not prefer heads over tails or vice versa,
i.e. distribution is symmetric.



Assume $a = 2$ and $b = 2$

$$\text{Beta}(\mu \mid a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1}$$



The coin tossing example: the Bayesian approach

Multiply likelihood by our prior ($l = n - m$):

$$p(\mu \mid m, l, a, b) = \underbrace{\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}}_{\text{constant } C} \mu^{a-1} (1-\mu)^{b-1} \prod_{i=1}^n \mu^{x_i} (1-\mu)^{1-x_i}$$

$$\log p(\mu \mid m, l, a, b) = \log(C) + (m+a-1) \log(\mu) + (l+b-1) \log(1-\mu)$$

$$\frac{\partial}{\partial \mu} \log p(\mu \mid m, l, a, b) = \frac{m+a-1}{\mu} - \frac{l+b-1}{1-\mu}$$



The coin tossing example: the Bayesian approach

Multiply likelihood by our prior ($l = n - m$):

$$\frac{\partial}{\partial \mu} \log p(\mu \mid m, l, a, b) = \frac{m + a - 1}{\mu} - \frac{l + b - 1}{1 - \mu}$$

$$\mu_{MAP} = \frac{m + a - 1}{n + a + b - 2}$$



The coin tossing example: the Bayesian approach

Now assume:

- Fair coin (i.e. true $\mu = 0.5$)
- Three coin tosses, heads 3x in a row (happens 1/8 times)

Central question:

What is the probability that the coin will land heads up?

$$p(x = 1|\mu) = \mu = \mu_{MAP}$$

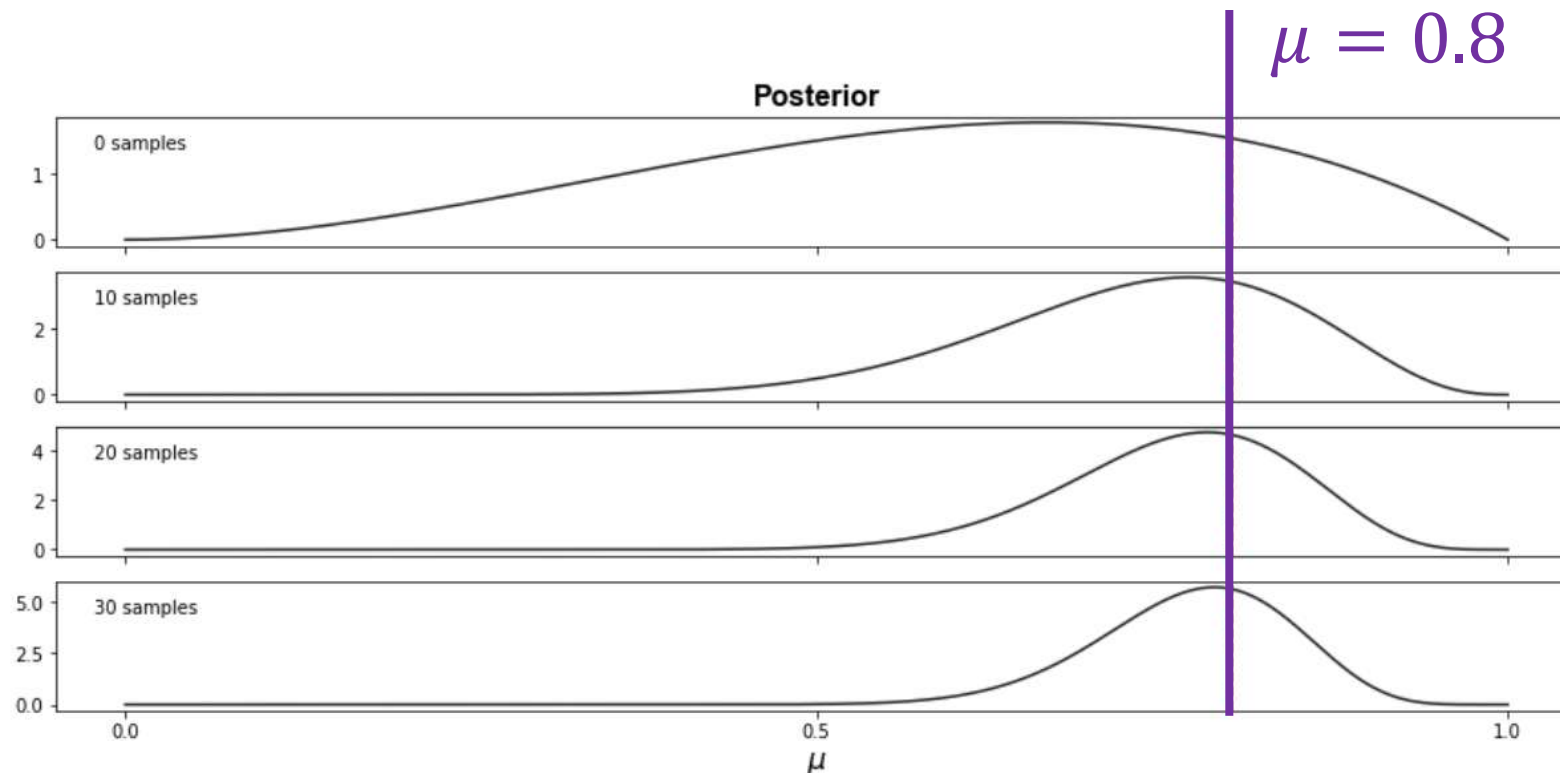
$$= \frac{m + a - 1}{n + a + b - 2} = \frac{3 + 2 - 1}{3 + 2 + 2 - 2} = \frac{4}{5}$$



The coin tossing example: the Bayesian approach

We can use μ_{MAP} to update our beliefs
(**sequential learning** based on increasing data)

After each data point, the posterior becomes the new prior

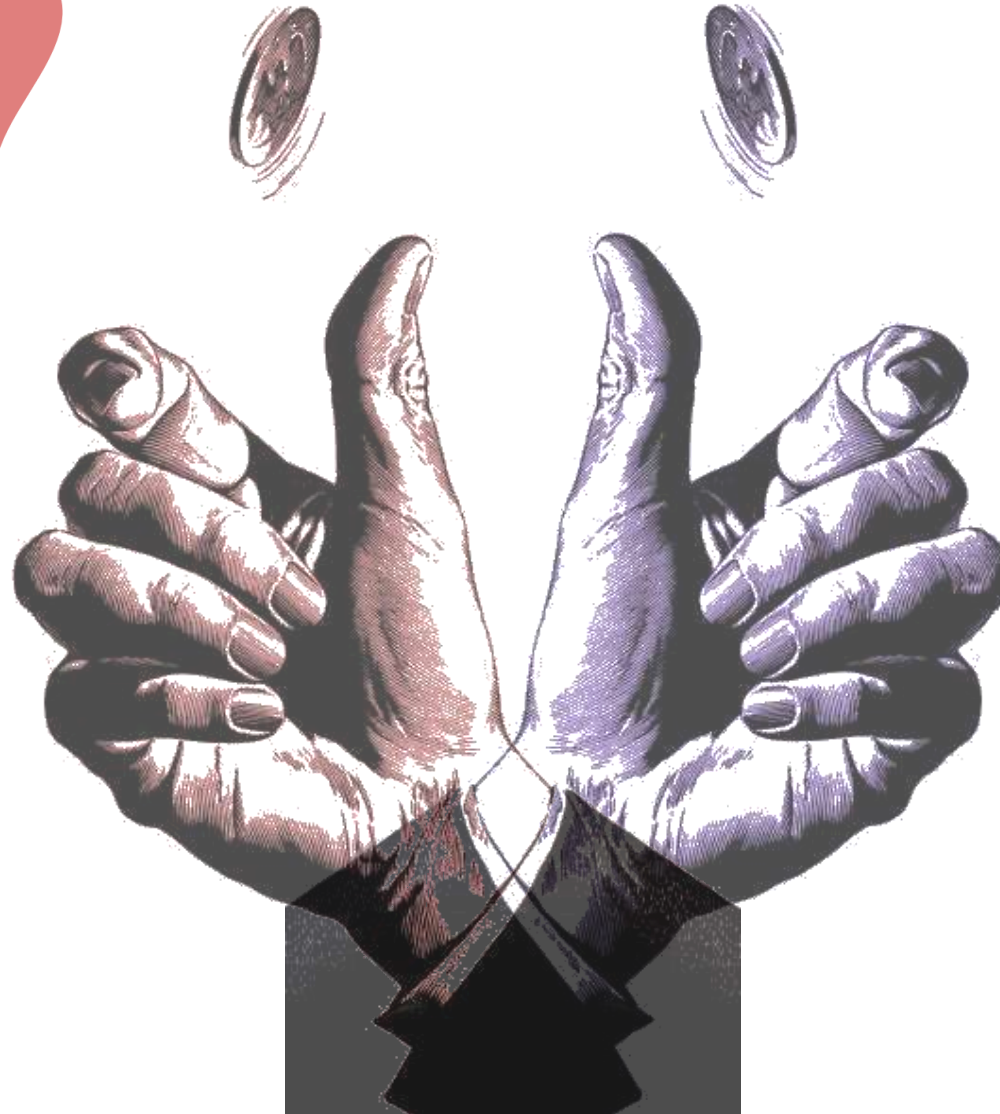


Frequentist vs Bayesian philosophy for solving inverse problems

FREQUENTISTS

Probability = frequency

- › Only repeatable events have probabilities
- › Probability of an event in a large number of trials > long-run frequency > requires repeatable experiments
- › QoIs are fixed
- › Predictions do not account for uncertainty
- › Optimization constraints affect interpretability of point estimates and confidence intervals



BAYESIANS

Probability = degree of belief

- › We can attach a probability to events, but also to beliefs > QoIs have a probability
- › Incorporate prior knowledge into analysis and update it with measurements
- › Predictions can account for uncertainty
- › Constraints are provided by prior and leave posterior to be interpreted as usual

Frequentist vs Bayesian philosophy: so what's best?

- › Mainly philosophical differences
 - › Data availability
- › Incorporation of prior belief

FREQUENTISTS

Probability = frequency

- › Generally, computationally more favorable
- › Requires large amounts of data to work well



BAYESIANS

Probability = degree of belief

- › Can incorporate prior beliefs
- › Priors can help when less data is available
- › Also works well with a lot of data
- › Up front about your assumptions

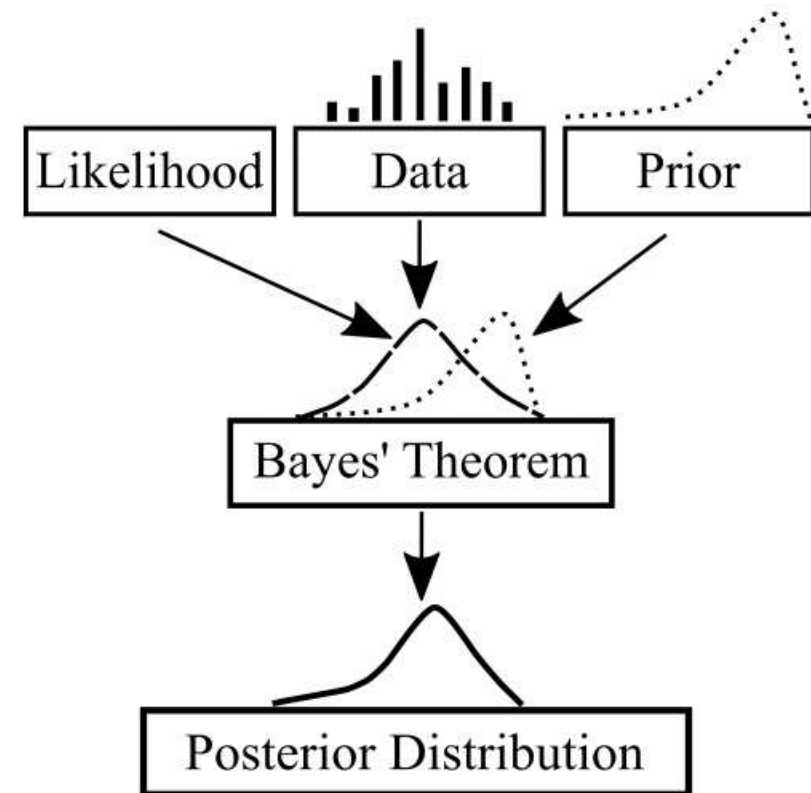
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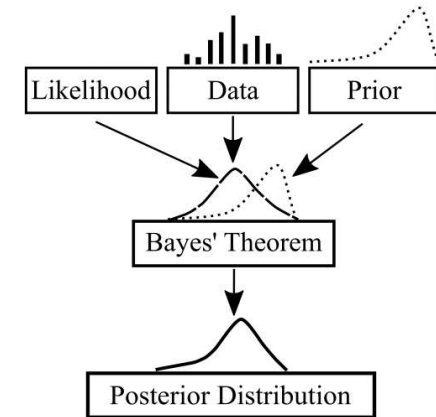


Bayesian philosophy

Bayes' theorem

describes the probability of a hypothesis/parameters, based on prior knowledge and observed evidence

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}$$



$p(\theta|x)$ **POSTERIOR:** probability of the hypothesis/parameter θ given the evidence x

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$p(x)$ **EVIDENCE:** probability of the evidence x arising, globally

Thomas Bayes [1701-1761]
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- Analytically
- Markov Chain Monte Carlo
- Variational Inference
- ...

ANALYTICAL BAYESIAN INFERENCE

Need for conjugacy?

In Bayesian inference, a prior $p(\theta)$ is **conjugate** to the likelihood function $p(x|\theta)$ when the posterior has the same functional form as the prior.

This means that the two boxed terms in Bayes' formula have the same functional form:

$$\boxed{p(\theta | x)} = \frac{p(x | \theta) \boxed{p(\theta)}}{\int p(x | \theta') p(\theta') d\theta'}$$

We already saw an example!

Conjugacy in Bayesian inference

We already saw an example!



PRIOR

$$\text{Beta}(\mu \mid a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1}$$

LIKELIHOOD

$$p(x_i = m) = \binom{n}{m} \mu^m (1-\mu)^{n-m}$$

POSTERIOR

$$p(\mu \mid m, l, a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \mu^{a-1} (1-\mu)^{b-1} \prod_{i=1}^n \mu^{x_i} (1-\mu)^{1-x_i}$$

$$\propto \mu^{m+a-1} (1-\mu)^{l+b-1}$$

conjugacy

Conjugacy in Bayesian inference



Some other examples:

- Beta-Binomial
- Beta-Bernoulli
- Gamma-Poisson
- Dirichlet-Multinomial
- ...



- Normal-Normal
- InverseGamma-Normal
- MultivarNormal-MultivarNormal
- InverseGamma-Weibull
- Gamma-Gamma
- ...

Major benefit:
posterior distribution can
be solved in close form

Conjugacy in Bayesian inference

$$p(\theta | x) = \frac{p(x | \theta) p(\theta)}{\int p(x | \theta') p(\theta') d\theta'}$$

Major benefit:

posterior distribution can
be solved in closed form

Unfortunately, only very few problems
can be solved using conjugate pairs

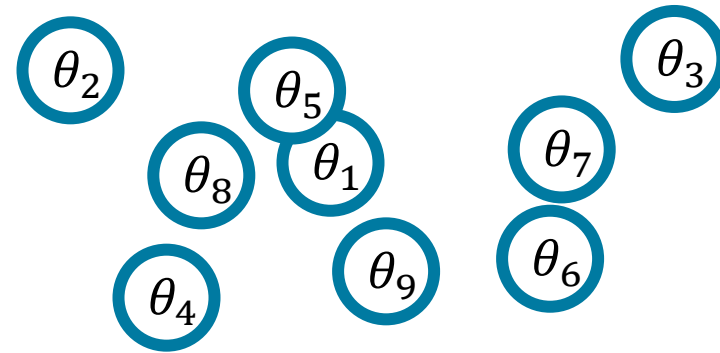
Need for approximation techniques, e.g.

- › **Markov Chain Monte Carlo**
- › **Variational inference**

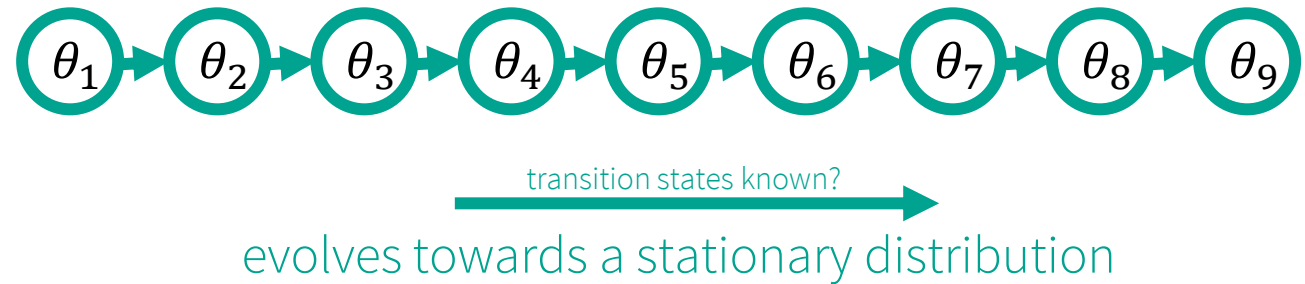
MARKOV CHAIN MONTE CARLO

Markov Chain Monte Carlo

Monte Carlo
independent sampling

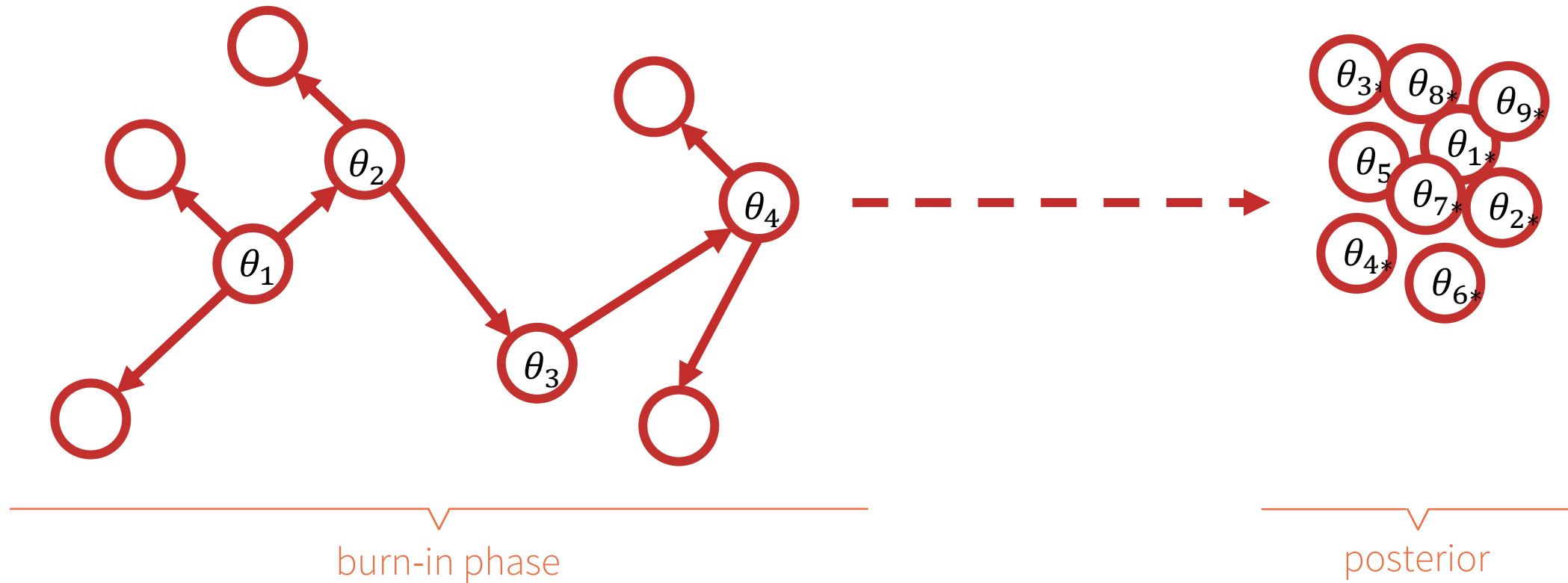


Markov Chain
next sample depends
on previous sample



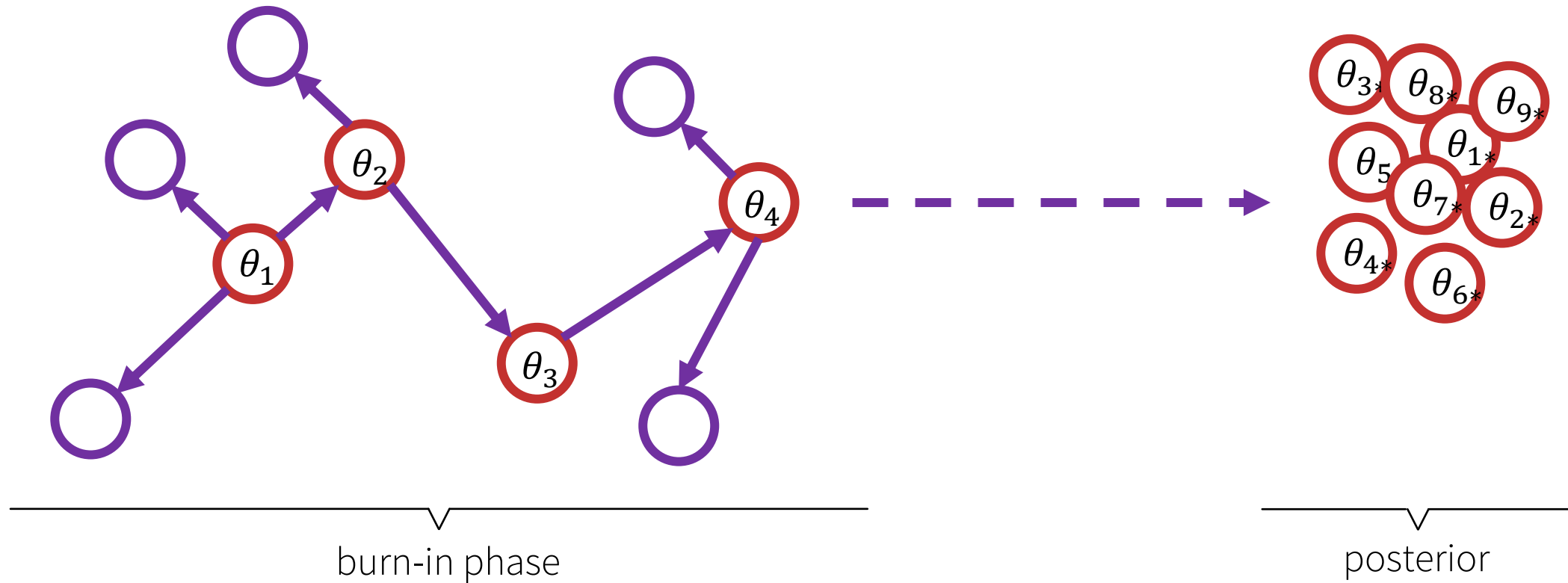
Markov Chain Monte Carlo

engineer a Markov Chain whose stationary distribution is the posterior distribution we are looking for



Markov Chain Monte Carlo

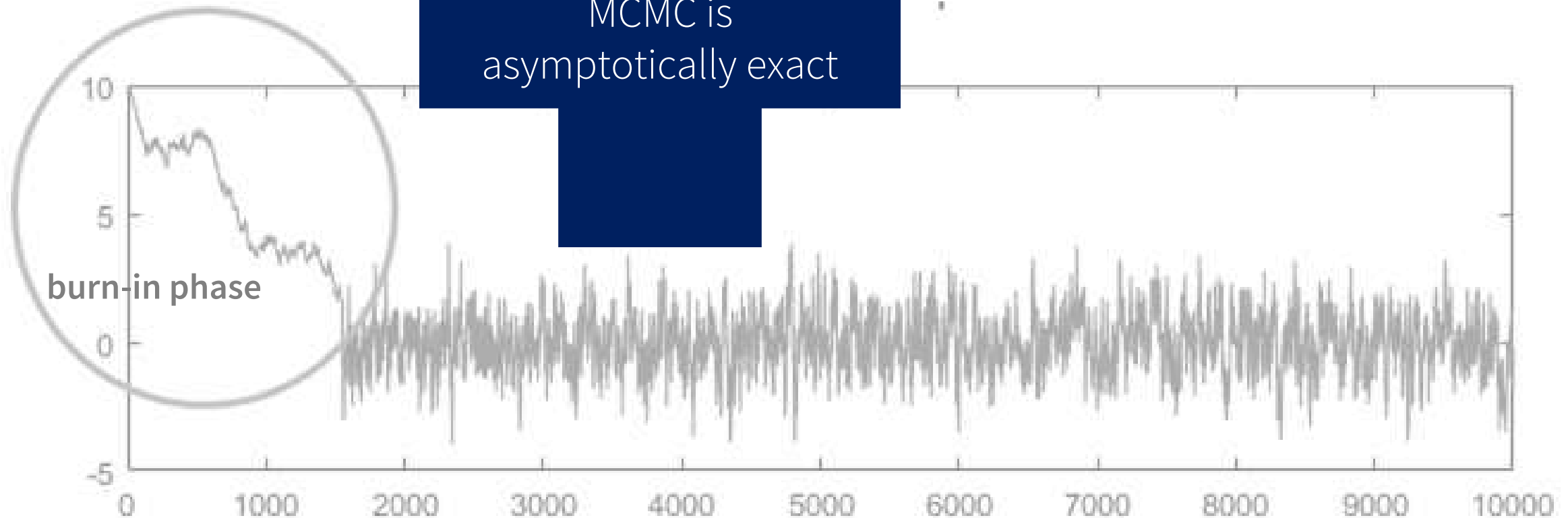
engineer a Markov Chain whose stationary distribution is the posterior distribution we are looking for



Markov Chain Monte Carlo

i.e. in the limit, MCMC will exactly approximate the target distribution

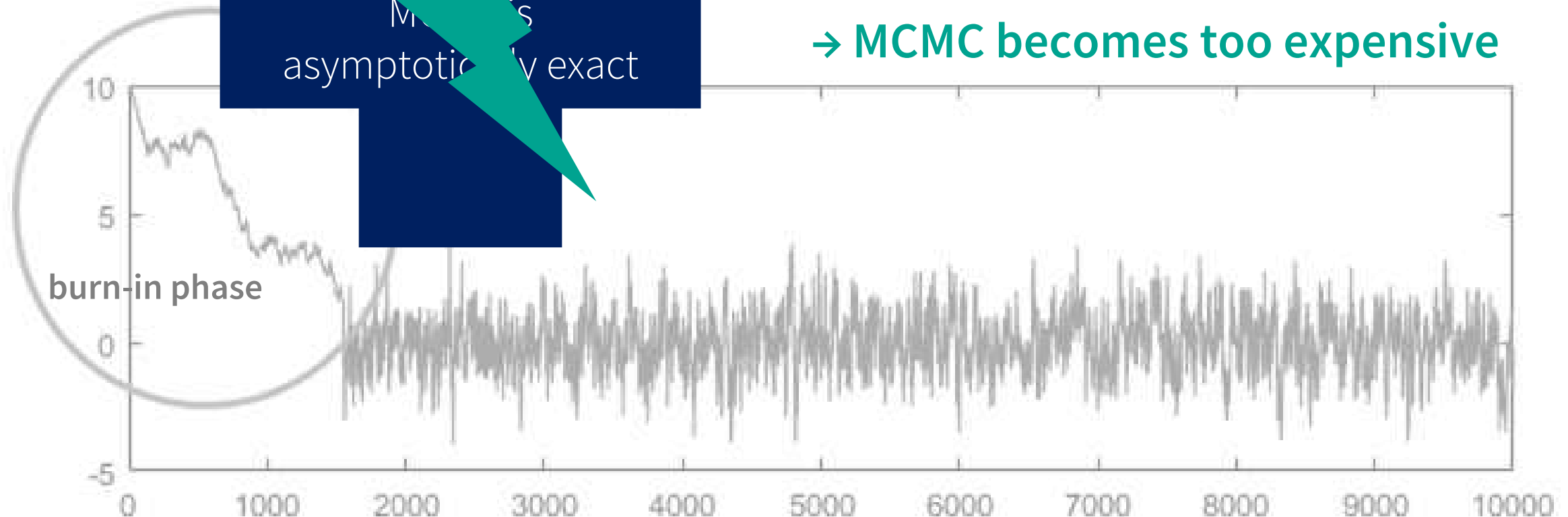
Major benefit:
MCMC is
asymptotically exact



Markov Chain Monte Carlo

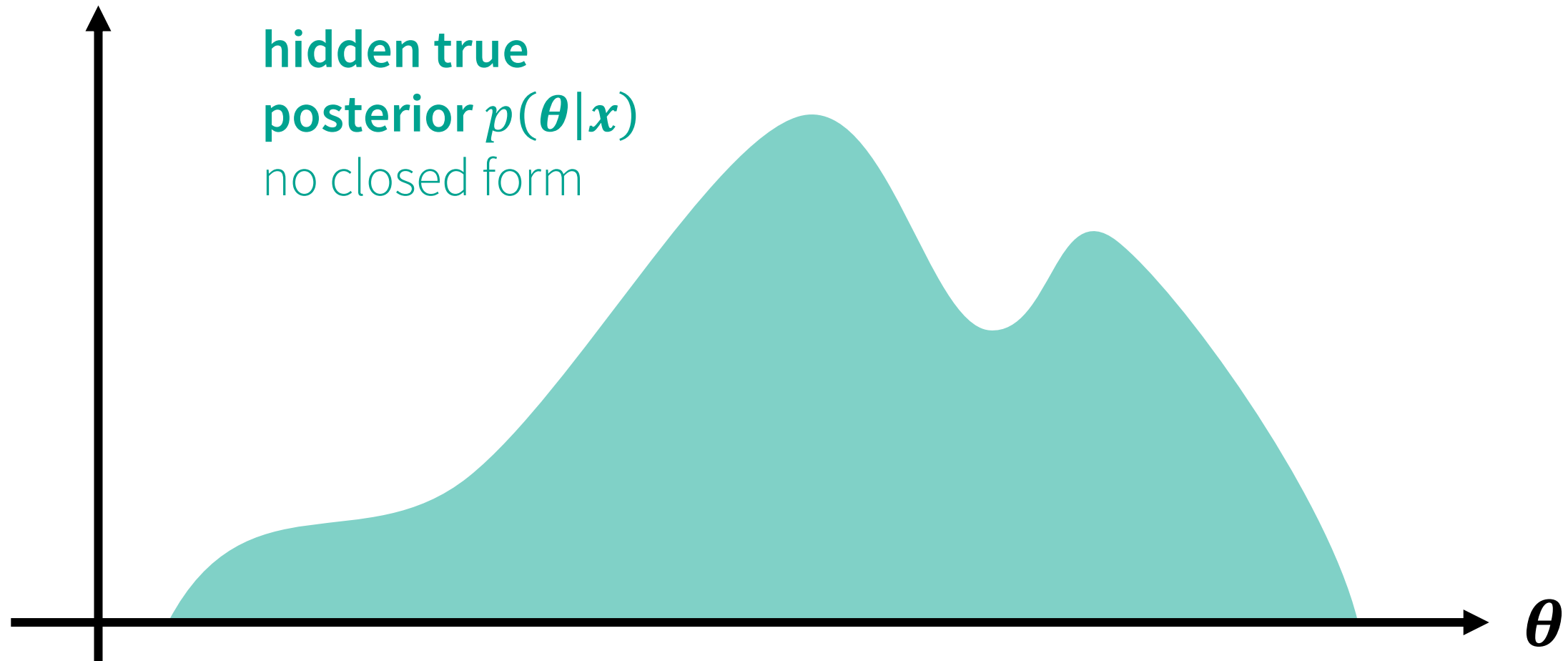
Major benefit:
MCMC is
asymptotically exact

- › large data
 - › many model features / parameters to infer
 - › slow sampling
- MCMC becomes too expensive

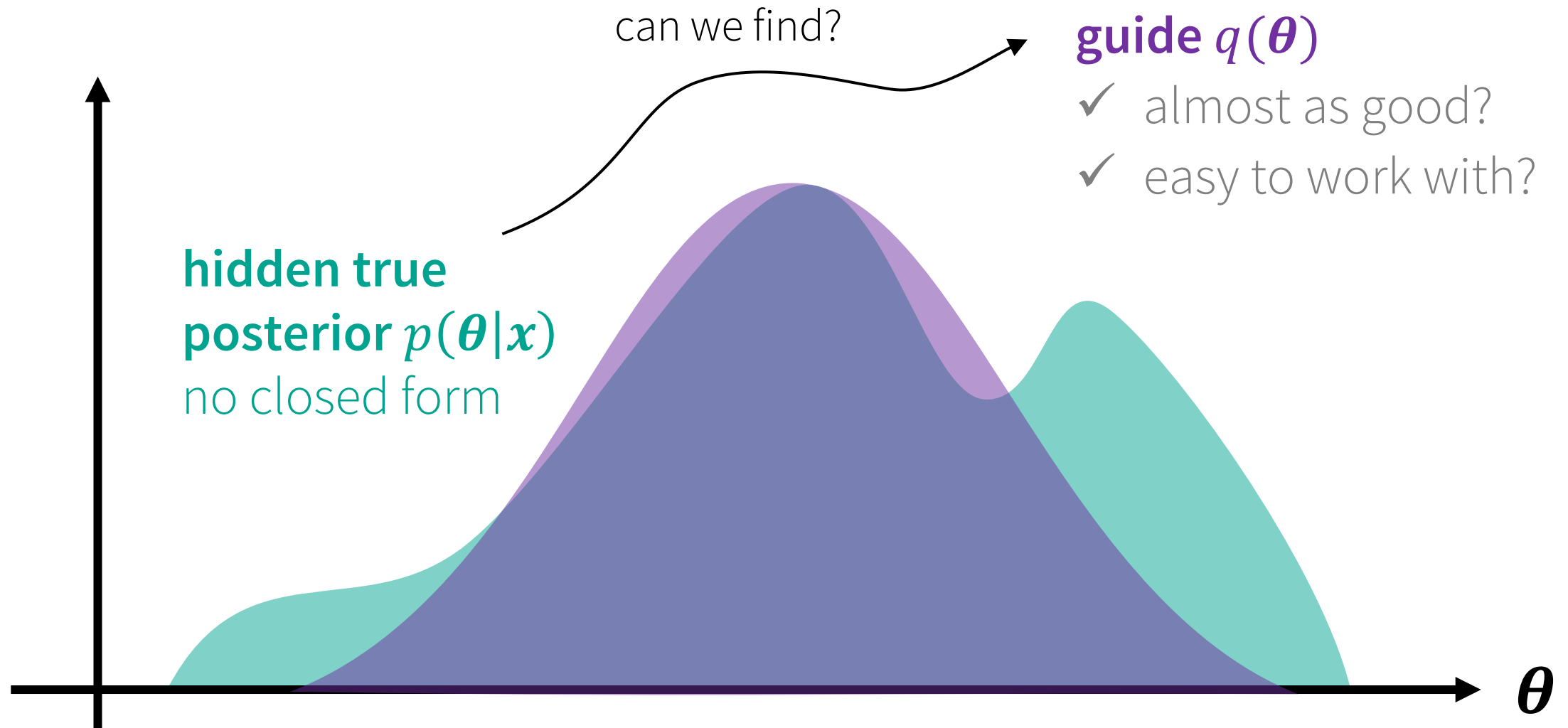


VARIATIONAL INFERENCE

Variational Inference



Variational Inference

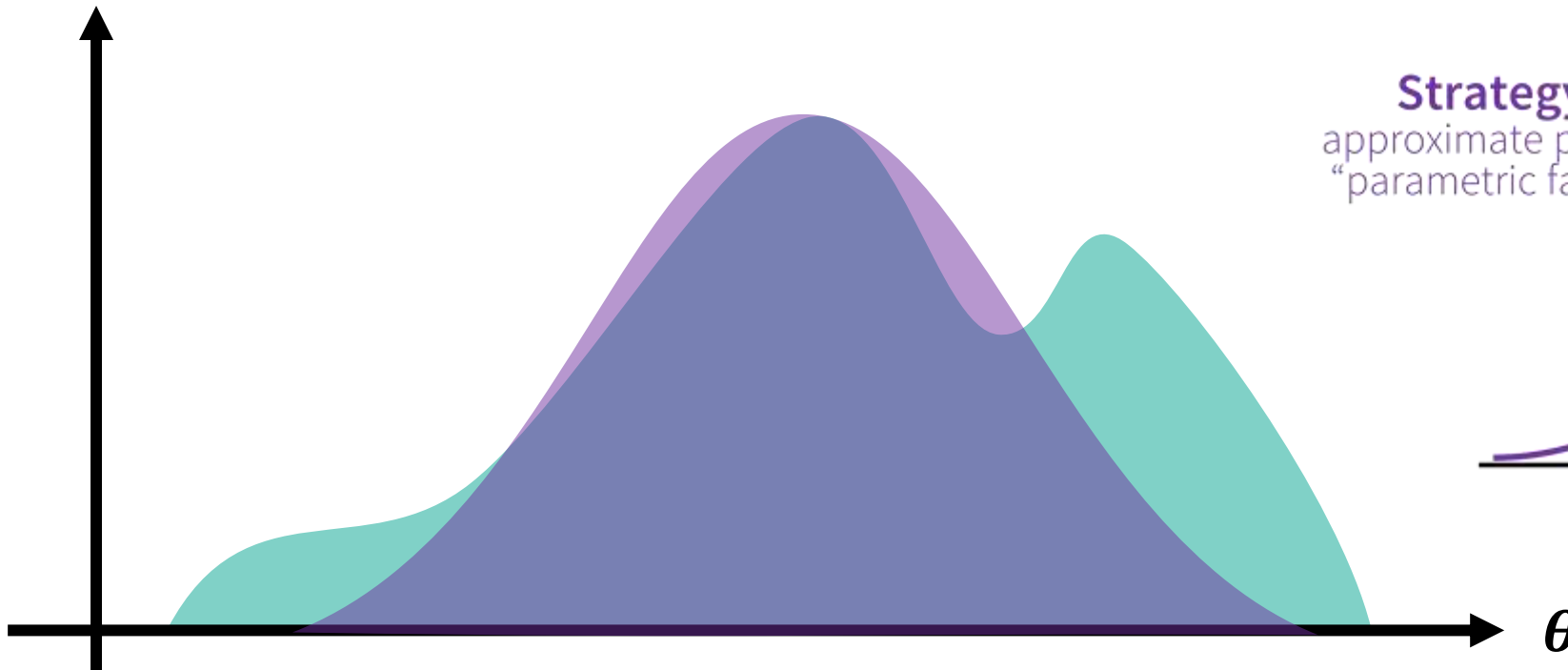


Variational Inference

in other words:

guide $q(\theta; \varphi) \approx$ **posterior** $p(\theta|x)$

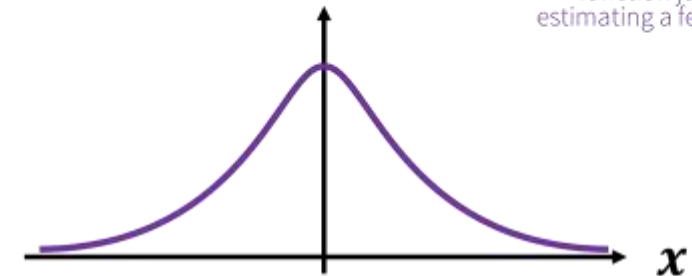
- ✓ almost as good?
- ✓ easy to work with?



Strategy:
approximate pdfs as
“parametric family”

$$p(x) \longrightarrow p_{\varphi}(x)$$

You can get an entire
function just from
estimating a few values φ

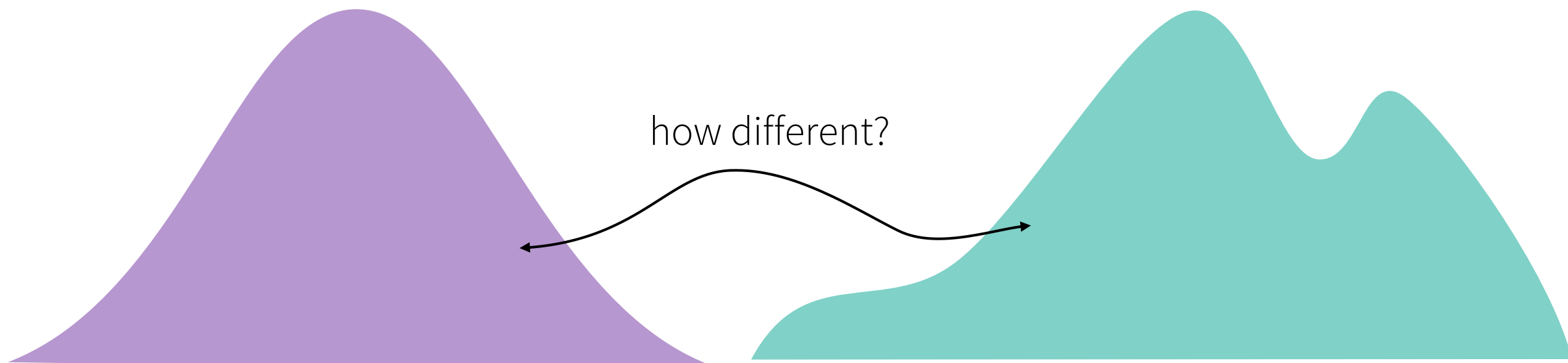


Variational Inference

guide $q(\theta; \varphi) \approx$ posterior $p(\theta|x)$

“variational calculus” problem:
we want to find the optimal for a function

**we need a loss metric that quantifies
the difference between two distributions**



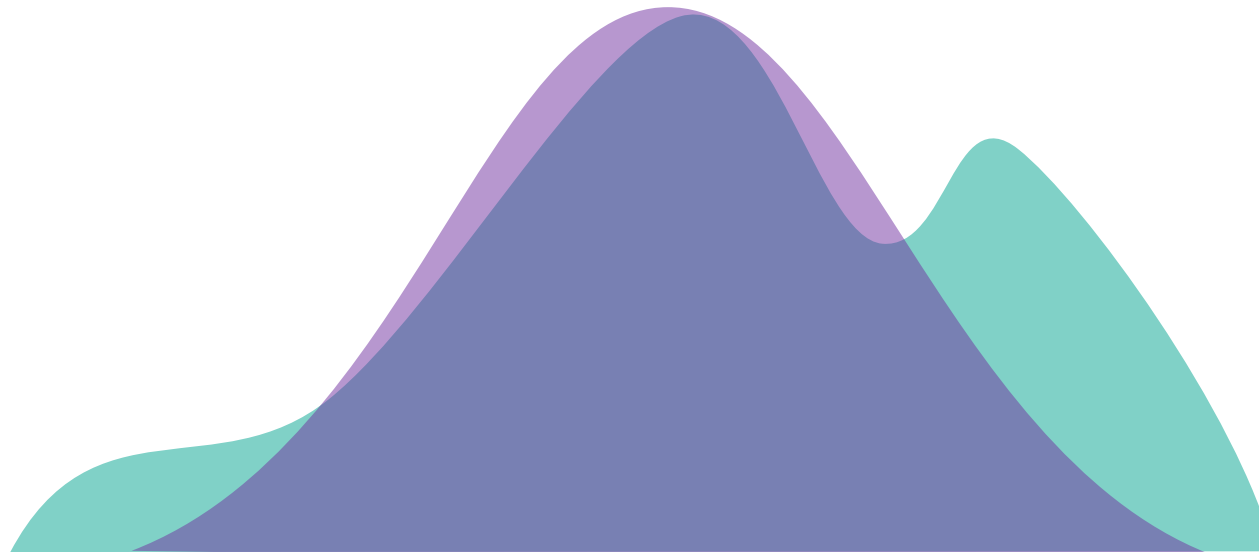
Variational Inference

$$\text{guide } q(\theta; \varphi) \approx \text{posterior } p(\theta|x)$$

we need a loss metric that quantifies
the difference between two distributions

Kullback-Leibler divergence

$$KL[q(\theta; \varphi) || p(\theta|y)]$$



Variational Inference

guide $q(\theta; \varphi) \approx$ posterior $p(\theta|x)$

Kullback-Leibler divergence

$$KL[q(\theta; \varphi) || p(\theta|x)] = \mathbb{E}_{q(\theta; \varphi)} \left[\log \left(\frac{q(\theta; \varphi)}{p(\theta|x)} \right) \right] = \int q(\theta; \varphi) \log \left(\frac{q(\theta; \varphi)}{p(\theta|x)} \right) d\theta$$

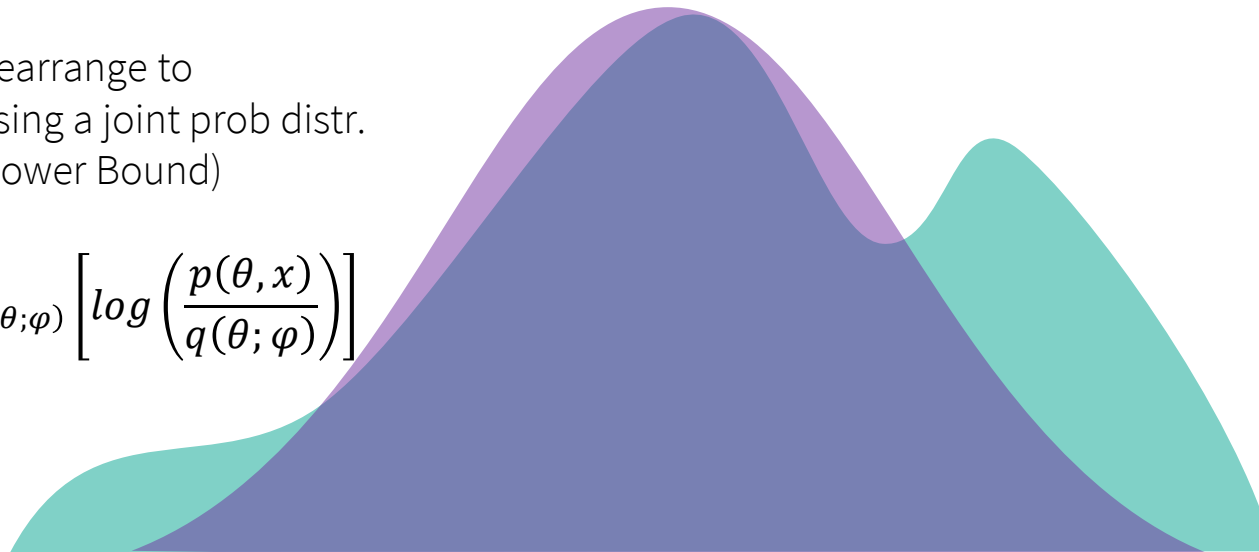


which we can rearrange to

$KL = -ELBO + \log p(x)$ using a joint prob distr.

ELBO (Evidence Lower Bound)

$$ELBO \mathcal{L}(q(\theta; \varphi)) = \mathbb{E}_{q(\theta; \varphi)} \left[\log \left(\frac{p(\theta, x)}{q(\theta; \varphi)} \right) \right]$$



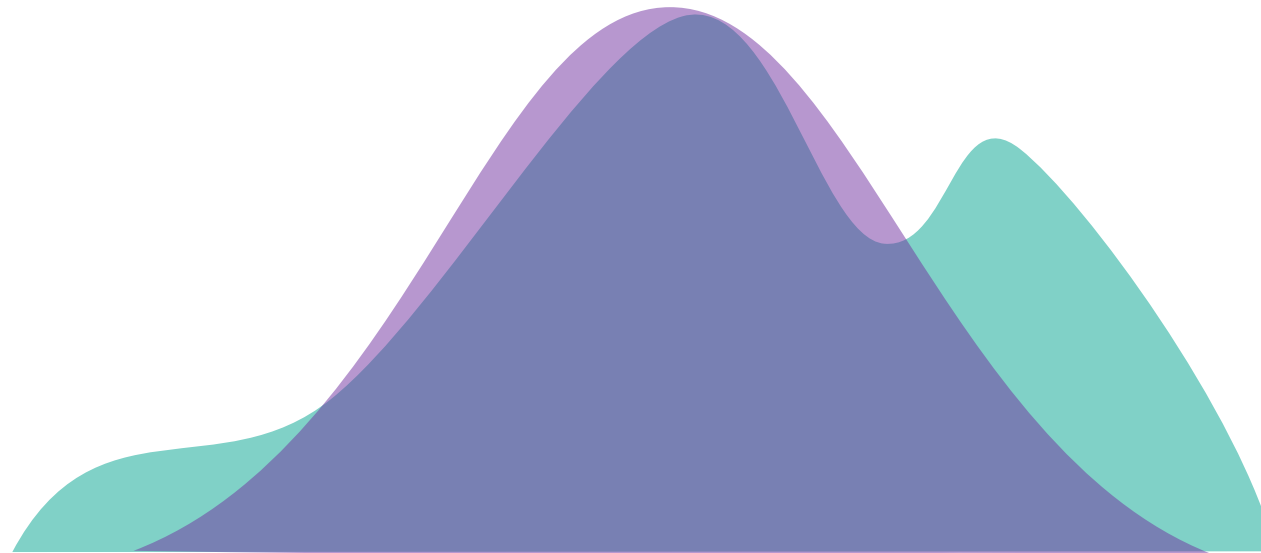
Variational Inference

$$\text{guide } q(\theta; \varphi) \approx \text{posterior } p(\theta|x)$$

we minimize the
KL divergence* to find
the most optimal guide function

$$\operatorname{argmin}_{q(\theta)} KL[q(\theta; \varphi) || p(\theta|y)] = \varphi^*$$

*i.e. we maximize the ELBO (Evidence Lower Bound)



Variational Inference: illustrative 1D example

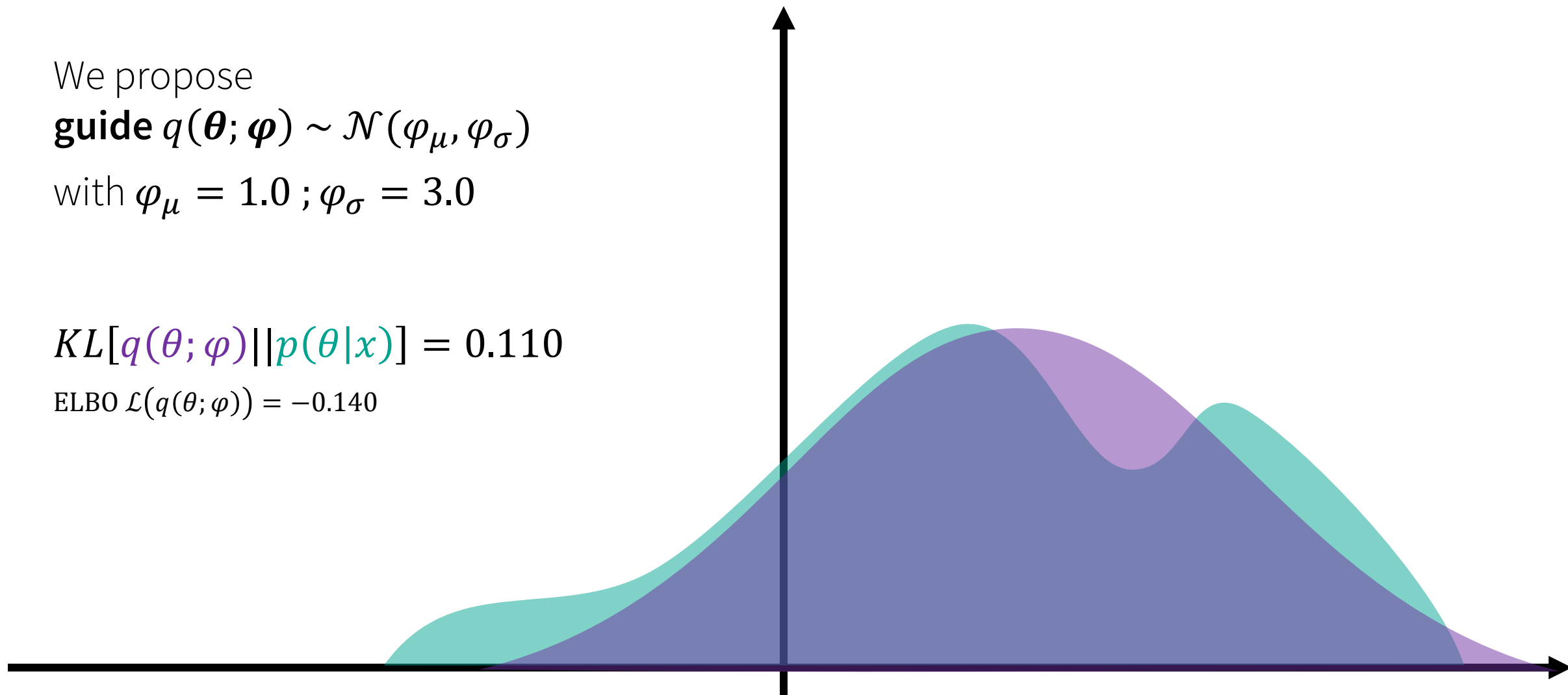
We propose

guide $q(\theta; \varphi) \sim \mathcal{N}(\varphi_\mu, \varphi_\sigma)$

with $\varphi_\mu = 1.0$; $\varphi_\sigma = 3.0$

$$KL[q(\theta; \varphi) || p(\theta|x)] = 0.110$$

$$\text{ELBO } \mathcal{L}(q(\theta; \varphi)) = -0.140$$



Variational Inference: illustrative 1D example

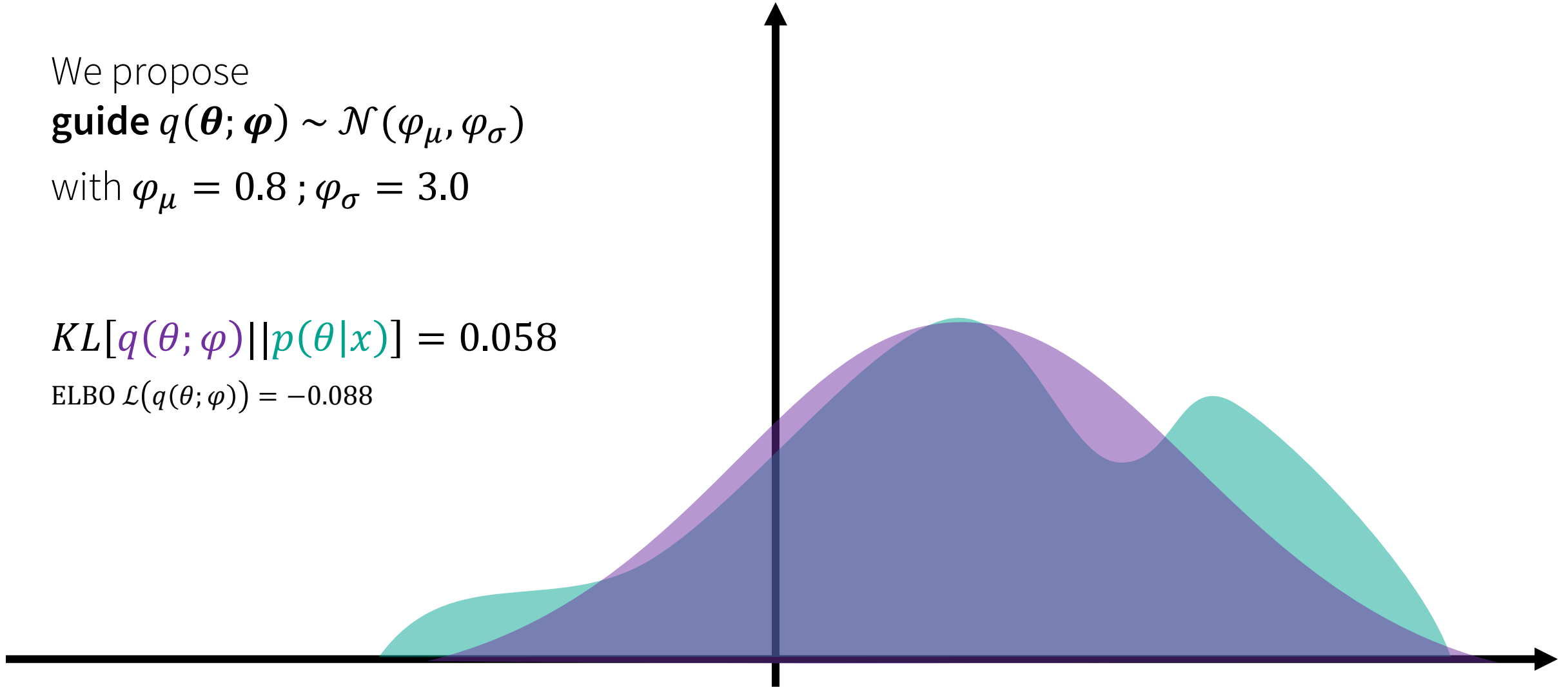
We propose

guide $q(\theta; \varphi) \sim \mathcal{N}(\varphi_\mu, \varphi_\sigma)$

with $\varphi_\mu = 0.8$; $\varphi_\sigma = 3.0$

$$KL[q(\theta; \varphi) || p(\theta|x)] = 0.058$$

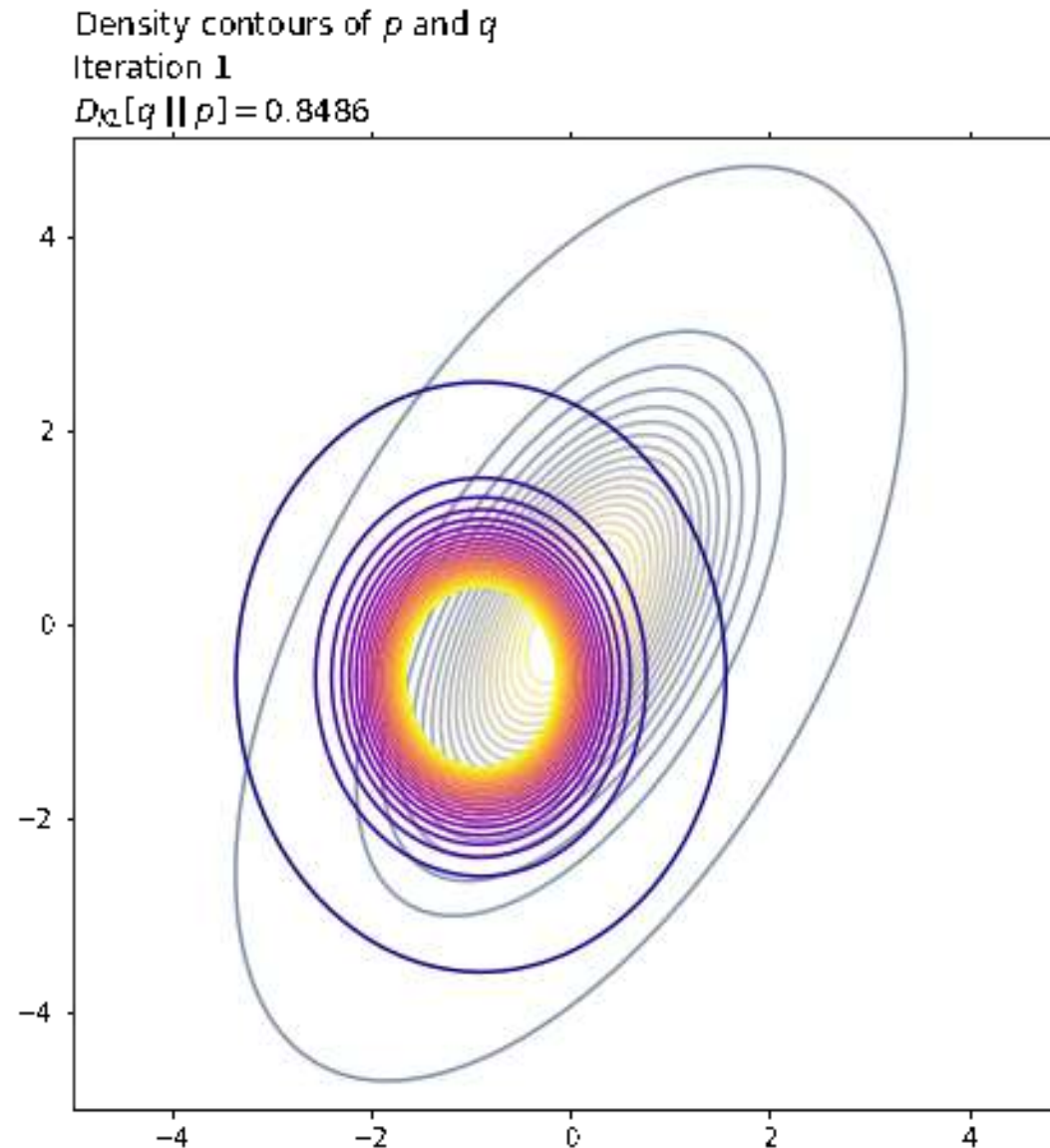
$$\text{ELBO } \mathcal{L}(q(\theta; \varphi)) = -0.088$$



Variational Inference: illustrative 2D example

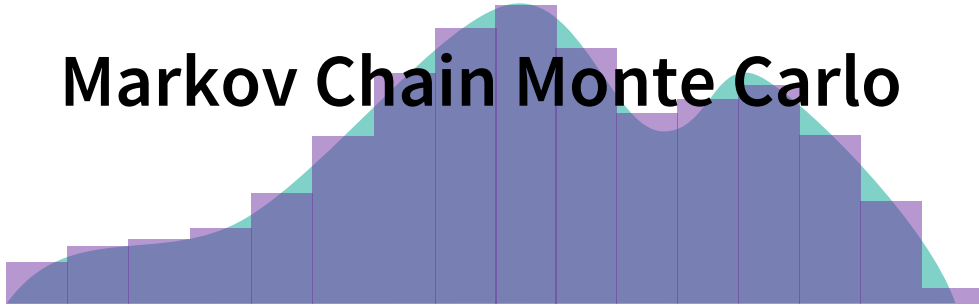
minimizing KL
shifts the proposal

minimum
determined by
shape of proposal



Comparison MCMC and VI

Markov Chain Monte Carlo



- › Sampling based
- › Burn-in phase → asymptotically exact samples from the true posterior
- › Sampling requires acceptance/rejection
- › Computationally more expensive
- › When to use:
 - Few datapoints which are very accurate
 - Precision is crucial

Variational Inference



- › Optimization based
- › Proposal/guide distribution is minimized w.r.t. the posterior
- › Guide is not the true posterior
- › Computationally more efficient
- › When to use:
 - Lots of data
 - Many model features

STILL UNCERTAIN?

we have a workshop!

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WORKSHOP

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